

Learn, Denoise and Discover: A Guide to Deep Denoising with an Application to Electron Microscopy

Sreyas Mohan*, Kangning Liu*, Peter A. Crozier, Carlos Fernandez-Granda

Abstract—This tutorial provides a self-contained description of deep-learning methodology for denoising, emphasizing aspects that are important in real-world scientific applications. The tutorial presents convolutional neural networks from first principles, explaining how to train them in a supervised and unsupervised manner. It also describes how to analyze the denoising strategies learned by these models, and how to evaluate their performance. Illustrative examples are provided based on computational experiments with simple 1D piecewise-constant signals, natural images and simulated electron-microscopy data. Code to reproduce all experiments is available online at <https://kangningthu.github.io/AI-powered-denoising>. In addition, a detailed case study is used to demonstrate the potential and challenges of applying deep denoising in practical scenarios. In the case study, supervised, unsupervised and semi-supervised deep-learning models are leveraged to denoise transmission-electron-microscopy data acquired at a very low signal-to-noise-ratio with the goal of investigating the dynamics of catalytic nanoparticles.

Index Terms—Denoising, deep learning, convolutional neural networks, scientific imaging, unsupervised learning, electron microscopy.

I. INTRODUCTION

Imaging technology is an essential tool for scientific discovery. Electron microscopy enables us to visualize the atomic structure of materials and viruses [127], fluorescence microscopy allows us to study cellular processes [66], and telescopes reveal astronomical objects that are light years away [75]. A fundamental challenge in imaging is the presence of stochastic perturbations that corrupt the acquired data. These perturbations, known as *noise*, often obscure key information. It is therefore often of critical importance in practice to *denoise* imaging data in order to facilitate scientific analysis.

In the literature, noise is often modeled as an independent, identically distributed random perturbation following a Gaussian distribution with fixed variance. However, in imaging applications such as microscopy, the dominant perturbations are often caused by shot noise, which follows Poisson statistics and is not independent of the signal amplitude [16]. A mixture of these two types of noise can be used to model perturbations in raw data from digital imaging sensors [36] and fluorescence microscopy [123]. Other important noise models include impulse noise [71] (sometimes known as salt-and-pepper noise), which produces sparse, impulsive perturbations, and non-additive noise processes such as speckle noise, which has a multiplicative effect on the signal of interest [4, 94].

Over the past decade, deep learning has revolutionized signal processing [63], and denoising is no exception. Deep neural networks are currently the state of the art for denoising on standard natural-image datasets [12, 120, 122, 19, 67, 52]. Deep denoisers are also the foundation of generative diffusion models [47, 91], as they are able to capture rich image statistics, which can be leveraged to generate high-quality synthetic images. However, the application of deep denoising to real-world data is not necessarily straightforward. The goal of this tutorial is to provide a self-contained description of deep-learning denoising methodology, emphasizing aspects that are important for its application to real imaging data. The tutorial is complementary to recent survey papers, which provide overviews of the growing literature on this topic [104, 105, 50, 30, 34, 42, 41, 53, 17, 2].

In Section II we provide a statistical formulation of the denoising problem. Section III describes denoising methodology developed prior to the advent of deep learning, including linear shift-invariant filtering, nonlinear filtering, and sparsity-based methods. In Section IV we present the most popular deep-learning architecture for denoising: convolutional neural networks (CNNs). Section V explains how to train these models in a supervised manner, using a database of noise-free and noisy signal pairs. In Section VI we investigate what strategies are learned by CNNs to denoise different types of data. Section VII shows that denoising CNNs can be trained in an unsupervised fashion, using exclusively noisy data. Section VIII discusses how to evaluate deep denoising methods.

In Section IX we apply supervised and unsupervised deep denoising to real-world data acquired via transmission electron microscopy (TEM). TEM is a powerful and versatile microscopy modality that is used in materials science to study the atomic-level structure and composition of materials, such as catalysts or semiconductors [99, 103]. TEM has had a huge impact in structural biology, as recognized with the award of the 2017 Nobel Prize in Chemistry [5], and is widely applied in other domains, including virology [9], paleontology [128] and nanotechnology [110].

Here, we focus on the application of TEM to characterize catalytic systems, where chemical transformations are accompanied by dynamic, atomic-level structural rearrangements occurring over tens of milliseconds [61, 21]. Recent technological advances in TEM make it possible to image these dynamic events [35, 32], but the high frame rates result in severe degradation by shot noise. This cannot be mitigated by

*Equal Contribution

increasing the intensity of the incident electron beam, because it would damage the material. Deep denoising has emerged as a powerful tool to overcome the limited signal-to-noise ratio and visualize previously unseen atomic dynamics [20].

Our application to electron microscopy illustrates important challenges that arise when applying deep-learning denoisers in practice. A supervised approach following Section V is shown to require a careful design of the training dataset, based on expert knowledge of the imaging process. The unsupervised framework in Section VII addresses this shortcoming, but its performance is limited due to the small quantity of available data. This is resolved by combining both approaches within a semi-supervised framework.

Throughout the tutorial we provide illustrative examples based on computational experiments with three types of signals, each corrupted by a different type of noise: (1) One-dimensional piecewise-constant signals perturbed additively by independent identically distributed (i.i.d.) Gaussian noise with fixed variance, which are a pedagogical example of simple signals with a single informative feature. (2) Natural photographic images perturbed additively by i.i.d. Gaussian noise with variable variance, which are a standard benchmark in the recent denoising literature. (3) Simulated electron-microscopy images of platinum nanoparticles corrupted by Poisson noise, which mimic the real-world data in Section IX. A more detailed description of these datasets is provided in Appendices A, B and C. Code to reproduce our experiments is publicly available online at <https://kangningthu.github.io/AI-powered-denoising>.

Notation

Throughout the tutorial we use a tilde to indicate random variables or vectors (e.g. $\tilde{x}$, $\tilde{y}$) and no tilde to indicate deterministic variables (e.g. x , y).

II. THE DENOISING PROBLEM

Denoising is an inverse problem, where the goal is to recover a signal from noisy measurements. This is achieved via a denoising function or model $\mathcal{D}$, which maps a noisy input y to a denoised approximation of a corresponding noise-free signal x ,

$$\mathcal{D}(y) \approx x. \quad (1)$$

Denoisers are typically designed to process a certain type of signal (e.g. natural images) and noise (e.g. additive Gaussian or Poisson noise). We can model this probabilistically, by representing the noise-free signal and the noisy measurements as random vectors, denoted by $\tilde{y}$ and $\tilde{x}$, whose joint distribution captures the characteristics of the signals of interest. In that case, the accuracy of the denoiser can be evaluated by computing the mean squared error (MSE) between the random vectors,

$$\text{MSE} := \mathbb{E} \left[\|\tilde{x} - \mathcal{D}(\tilde{y})\|_2^2 \right], \quad (2)$$

or, equivalently, a logarithmic function of the MSE defined on a decibel scale called the peak signal-to-noise ratio (PSNR)

$$\text{PSNR} := 10 \log \left(\frac{M^2}{\text{MSE}} \right), \quad (3)$$

where M is a fixed constant representing the maximum possible value of the signal of interest. In practice, the MSE and PSNR are typically approximated via averaging on image databases consisting of noise-free and noisy pairs of images.

Mathematically, the optimal estimator in terms of MSE is the conditional mean $\mathbb{E}[\tilde{x}|\tilde{y}]$ [48]. Naively, one could think that denoising is a solved problem: we only need to compute the conditional mean of the noise-free signal given the noisy measurements. Unfortunately, this is easier said than done. Images are high-dimensional signals, as they typically consist of thousands or even millions of pixels. This makes it intractable to compute the conditional mean from the available data without making simplifying assumptions about the structure and distribution of the signals. Many denoising methods can be interpreted as approximating the conditional mean under such assumptions.

An important shortcoming of the MSE and PSNR is that they may not adequately reflect signal features that are important for specific applications. In the case of natural images, they tend to neglect textures and fine-scale details, which has motivated the development of perceptual metrics to assess the visual quality of natural images [124, 111, 121]. In fact, there exists a perception-distortion tradeoff, as established in [10], where optimizing the mean distortion of a denoised image deteriorates perceptual quality (see also [30]).

III. DENOISING BEFORE DEEP LEARNING

In this section we describe denoising methodology developed before the advent of deep learning. Perhaps the simplest of all denoisers is the linear shift-invariant (LSI) filter, which is still used in many practical applications to this day. When we input noisy measurements, represented by a deterministic vector y , into a filter that is linear and shift-invariant, the denoised signal $\mathcal{D}_{\text{LSI}}(y)$ is the *convolution* between y and the LSI filter w_{LSI}

$$\mathcal{D}_{\text{LSI}}(y) = w_{\text{LSI}} * y. \quad (4)$$

As explained in Section IV, convolutions are a key element of deep-learning methodology for denoising.

The ideal LSI filter that minimizes the MSE between the denoised estimate and the underlying noise-free signal is known as the Wiener filter [114]. If we represent the noise-free signal and the noisy data as the random vectors $\tilde{x}$ and $\tilde{y}$, the Wiener filter satisfies

$$w_{\text{Wiener}} := \arg \min_w \mathbb{E} \left[\|\tilde{x} - w * \tilde{y}\|_2^2 \right], \quad (5)$$

The denoising effect of LSI filters is best understood in the frequency domain. A 1D or 2D signal is mapped to the frequency domain by computing its discrete Fourier transform (DFT), which we denote by $\mathcal{F}$. By the convolution theorem, the DFT of a convolution between two signals is equal to the elementwise or Hadamard product of their DFTs. Moreover, the DFT is an orthogonal transformation that preserves the ℓ_2 norm (up to a constant factor). As explained in Appendix G, this implies that the MSE optimization problem in (5) decouples into the different frequency components, so that the k th DFT coefficient of the Wiener filter $\mathcal{F}w_{\text{Wiener}}[k]$ only depends

on the corresponding DFT coefficients $\mathcal{F}\tilde{x}[k]$ and $\mathcal{F}\tilde{y}[k]$ of the noise-free and noisy signal:

$$\mathcal{F}w_{\text{Wiener}}[k] := \arg \min_f \mathbb{E} \left[(\mathcal{F}\tilde{x}[k] - f \cdot \mathcal{F}\tilde{y}[k])^2 \right]. \quad (6)$$

The Wiener filter therefore scales each frequency component of the noisy data separately. For example, when the noise is i.i.d. Gaussian with fixed variance, the k th scaling coefficient equals

$$\mathcal{F}w_{\text{Wiener}}[k] = \frac{v_k}{v_k + \sigma^2}, \quad (7)$$

where v_k is the variance of the k th DFT coefficient of the signal and σ^2 is the variance of the noise in the frequency domain (see Section 5 in [13] for a derivation). As a result, the filter suppresses frequency components that have small variance with respect to the noise variance, and preserves those that have large variance. In practice, the true statistics of the signals of interest are not known, so the Wiener filter can only be approximated, either by estimating the relevant statistics from the available data, or by minimizing an approximation of the MSE.

Figure 1 shows the result of applying an LSI filter to our example datasets. The LSI filter is obtained by minimizing an approximation to the MSE on the training datasets described in Appendices A, B and C. The low-frequency components of the 1D piecewise-constant functions and the natural images capture most of their variance, so the LSI filter acts as a low-pass filter. As a result, it tends to eliminate high-frequency information associated with fine-scale features such as sharp edges. In contrast, the electron-microscopy images contain periodic arrangements of atoms, so much of their variance is concentrated in the corresponding frequency components. The LSI filter preserves these components, and shrinks the rest. This results in the appearance of periodic artifacts in the vacuum, and the suppression of boundary information that is critical to understand the atomic structure of the depicted nanoparticles.

Prior to deep-learning, multiple nonlinear methods were developed to address the limitations of LSI filtering. In contrast to LSI filters, these methods are not shift invariant; they adapt to local signal statistics. The bilateral filter performs a local average with weights that depend on the signal intensity [106]. The locally-adaptive regression kernel method leverages a kernel function to further tailor a local averaging operation [102]. Non-local means [11] performs a non-local average, which identifies relevant pixels by comparing patches across the noisy image. It is worth noting that many of these methods are able to perform *unsupervised* denoising, implicitly learning signal statistics directly from the noisy data [77], in the same spirit as the unsupervised deep denoisers presented in Section VII.

Sparsity-based methods are an important class of nonlinear denoising approaches, which map the noisy signal to a domain where the noise-free signal is sparse (meaning that its variance is concentrated in a small number of coefficients). Denoising is then performed by discarding components that are small relative to the noise level, via nonlinear-thresholding strategies [26, 96, 15], which can be learned from data [46, 88], or

via minimization of a sparsity-inducing cost function [18, 93]. Initially, multiscale transformations such as wavelets [72], curvelets [101], the steerable pyramid [97], and others were specifically designed to yield sparse representations of natural images. Subsequently, data-driven learning approaches were proposed, which can be divided into *synthesis* and *analysis* methods [31]. Synthesis methods minimize the reconstruction error in a dictionary of learned basis functions [29, 70]. Analysis methods learn transforms that sparsify the signals of interest, which yields comparable or even improved results at a reduced computational cost [89, 113, 90, 112, 85]. Deep neural networks learn how to denoise directly from data in a similar spirit to these approaches, but leverage more flexible nonlinear transformations with many more parameters.

Finally, the block-matching and 3D filtering (BM3D) method [23] deserves mention due to its strong performance on natural images. BM3D applies a collaborative-filtering strategy, which enforces joint sparsity of similar patches extracted from an image in a discrete-cosine-transform domain. Weighted nuclear norm minimization is a related approach that enforces low-rank structure instead of joint sparsity [43].

IV. CONVOLUTIONAL NEURAL NETWORKS

In this section we describe convolutional neural networks (CNNs), a deep-learning model that was originally developed for image classification [3, 62], but has become a powerful workhorse for data-driven signal denoising.

A. Basic structure

Deep neural networks implement complex, nonlinear functions by interleaving linear transformations learned from data with pointwise nonlinearities. In the case of CNNs, the linear transformations are implemented by *convolutional layers*. The initial layer $\mathcal{C}_1$, which is made of c filters, maps the input signal x to a hidden representation $h := \mathcal{C}_1 x$ via convolution with multiple filters. Each filter w_l yields a *channel*

$$h[l] := b_l + w_l * x, \quad (8)$$

where l ranges between 1 and the number of channels c , and b_l is an additive constant, known as a *bias* parameter. We assume that the convolutional filters preserve the spatial dimensions of the input for simplicity. The hidden representation h can be represented as a tensor of dimensions $d \times c$ if the input signal is 1D signal with d entries, or $d \times d \times c$ if the input signal is a $d \times d$ image. Figure 2 shows the hidden representations of a simple two-layer CNN applied to a 1D piecewise-constant signal.

The subsequent convolutional layers operate in a similar way to the initial layer. They receive an input h_{in} with multiple (c_{in}) channels, which they transform into an output h_{out} with a number of channels c_{out} that can be different from c_{in} . Each output channel is obtained by applying a different convolutional filter to each input channel and summing the result,

$$h_{\text{out}}[l] := b_l + \sum_{m=1}^{c_{\text{in}}} w_l^{[m]} * h_{\text{in}}[m], \quad 1 \leq l \leq c_{\text{out}}. \quad (9)$$

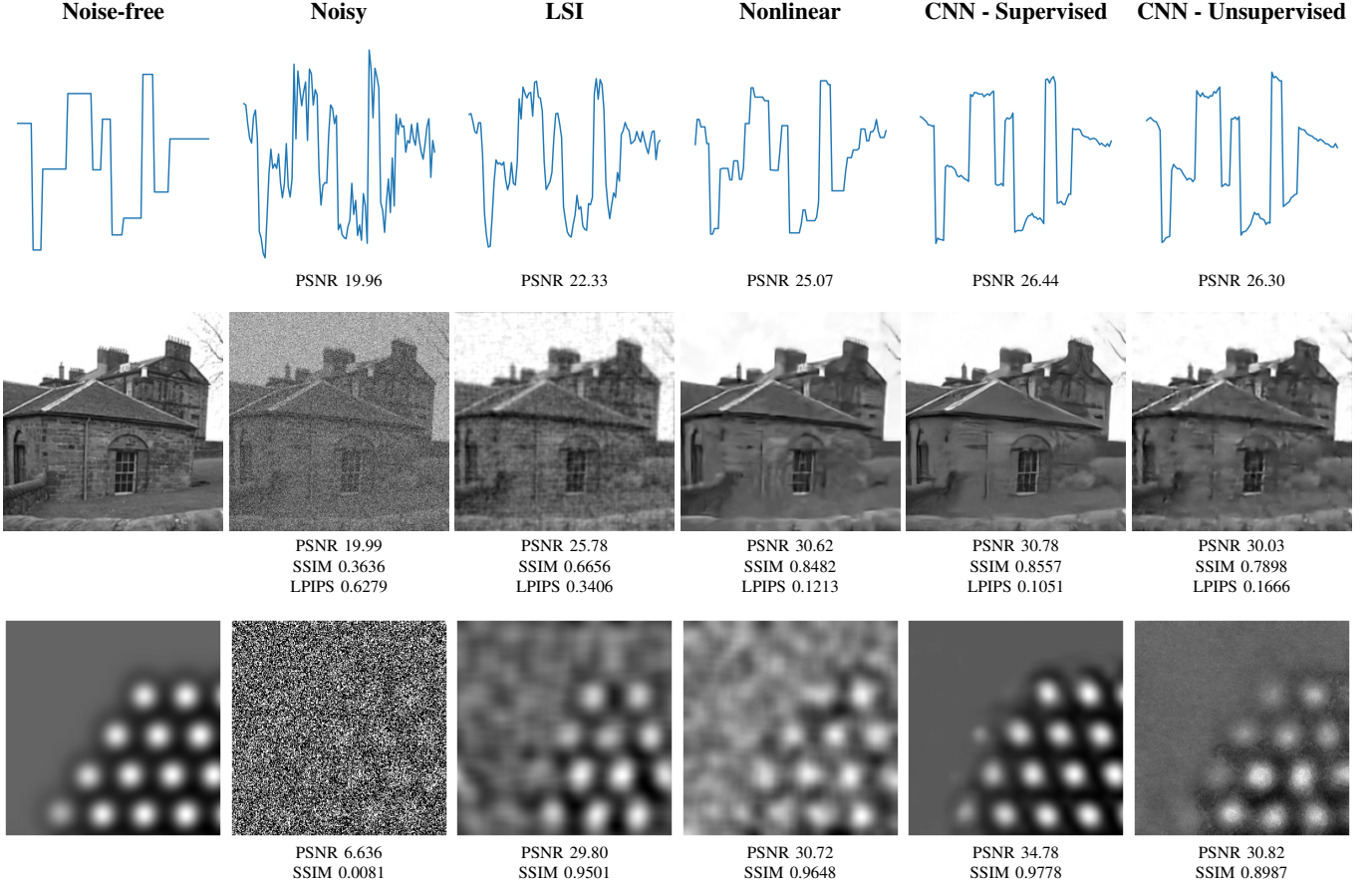


Fig. 1. **Examples and test denoising performance.** Examples showing noise-free, noisy and denoised signals from the three types of data used in our experiments: 1D piecewise-constant functions with additive Gaussian noise (first row), natural images with additive Gaussian noise of varying variance (second row) and electron-microscopy data with Poisson noise (third row). Four denoisers are compared: a linear shift-invariant filter (LSI) approximating a Wiener filter (see Section III), a nonlinear method not based on deep learning, a CNN trained in a supervised manner as described in Section V, and a CNN trained following the unsupervised Noise2Noise framework described in Section VII-B. The nonlinear baselines are total-variation regularization (first row), BM3D (second row) and PURE-LET (third row). The test metrics achieved by each denoiser in a held-out test set are reported below each example. The supervised CNN outperforms the other methods, although BM3D achieves similar results for natural images. The unsupervised CNN is close to that of the supervised CNN, except for the electron-microscopy data due to the limited number of training data (see Figure 12). Detailed descriptions of the datasets, CNN models and training procedure are provided in Appendices A, B, C, D and E.

Here $w_l^{[m]}$ denotes the m th convolutional filter and b_l the bias parameter associated with the l th channel. In total the convolutional layer has $c_{\text{in}} \times c_{\text{out}}$ filters, since c_{in} filters are applied to each input channel in order to generate each output channel. In the case of color images (or other imaging modalities with multiple values associated to each pixel), each color can be represented as a channel, so the initial layer also receives an input with multiple channels.

The output of each convolutional layer in a CNN (except the last one) is fed into a pointwise nonlinearity or *activation function*. This is crucial, as it allows the CNN to implement nonlinear functions. The most popular activation function is the rectified linear unit (ReLU) [38], which sets negative entries to zero and preserves nonnegative entries, as illustrated in Figure 2. Applying a ReLU layer to a hidden representation h_{in} yields the rectified representation $h_{\text{out}} := \mathcal{R}_+ h_{\text{in}}$, whose j th entry equals

$$h_{\text{out}}[l]_j := \begin{cases} h_{\text{in}}[l]_j & \text{if } h_{\text{in}}[l]_j \geq 0, \\ 0 & \text{otherwise.} \end{cases} \quad (10)$$

The rectified representation is typically called a *feature map*

or *activation map*. The ReLU can be applied very efficiently, and computing its derivative is trivial.¹ In addition, it provides superior performance to the sigmoid nonlinearities used prior to the 2010s, because its output does not saturate for large inputs [40]. Saturation is problematic because the gradient with respect to the parameters of the initial layers may become negligible, a phenomenon known as *vanishing gradients*, which makes it very challenging to learn the network parameters.

Figure 3 compares the denoising performance of CNNs with different number of convolutional layers for 1D piecewise-constant functions corrupted by additive Gaussian noise. Even a simple two-layer network like the one shown in Figure 2 outperforms the LSI filter, and achieves similar performance to total-variation regularization, which is an effective nonlinear sparsity-based method specifically designed for this kind of signal [14, 93]. Adding more layers further improves performance, clearly outperforming total-variation regularization. However, beyond 10 layers the improvement is very modest.

¹Strictly speaking, the ReLU is not differentiable when its input is exactly equal to zero, but this is not problematic in practice. The derivative can be set to zero, which is a subderivative of the function at that point.

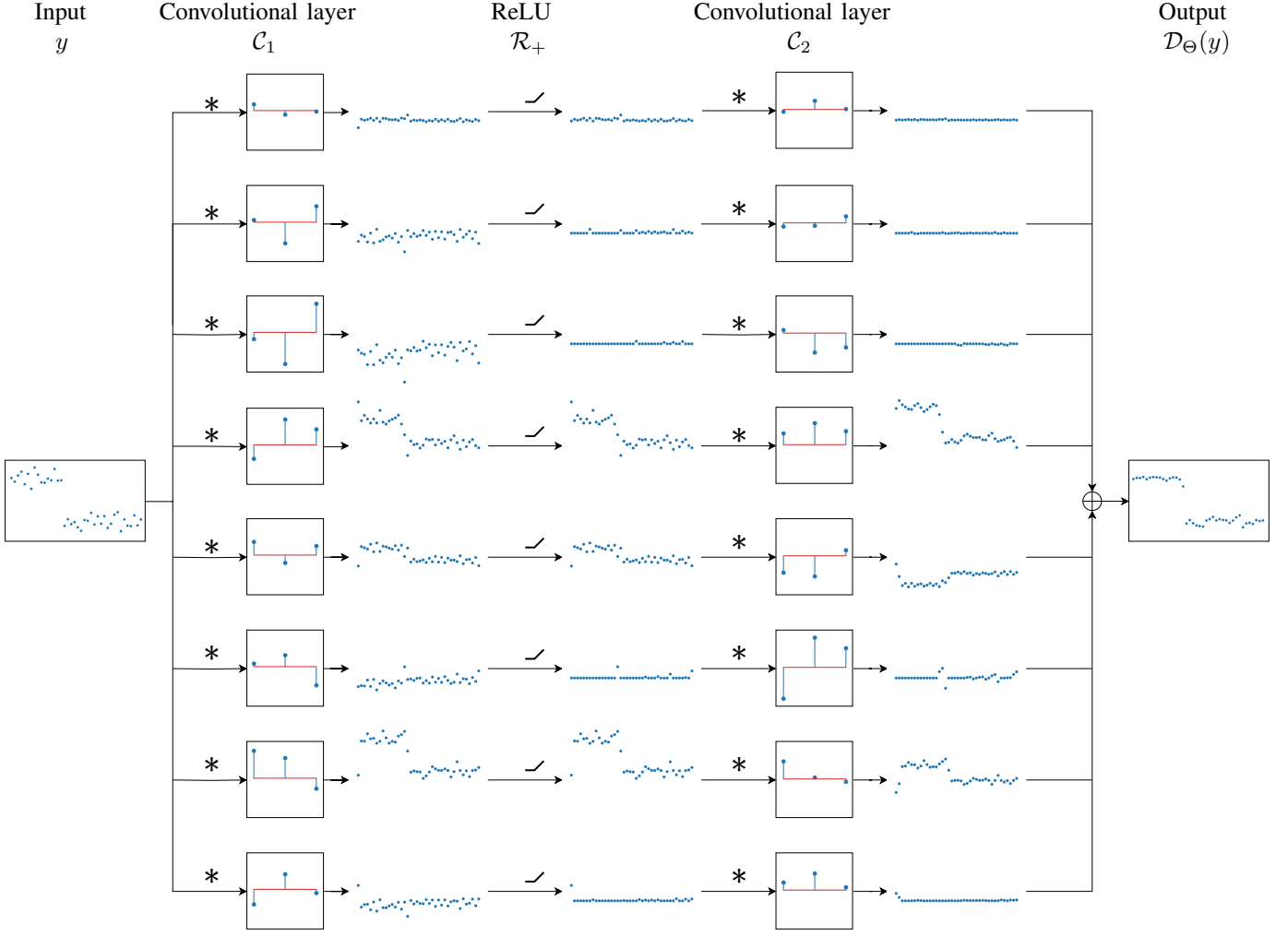


Fig. 2. **Convolutional neural network with two layers.** The diagram shows how a simple convolutional neural network denoises a one-dimensional piecewise-constant signal. The noisy input y is convolved with eight different filters in the first convolutional layer $\mathcal{C}_1$ to produce a hidden representation with eight channels. The representation is then fed into a ReLU nonlinearity $\mathcal{R}_+$, which yields the corresponding feature map by setting negative entries to zero. Finally, each channel is convolved with another convolutional filter in the second convolutional layer $\mathcal{C}_2$. The convolved outputs are summed together to produce the denoised output $\mathcal{D}_{\Theta}(y)$. The learnable parameters Θ are the 16 convolutional filters (for simplicity, the convolutional layers do not include additive bias parameters).

This is a typical phenomenon in the design of neural-network architecture: as long as the model has enough complexity to capture the relevant signal features, the specific number of layers does not make a big difference.

B. Batch normalization and skip connections

In order to train deep CNNs effectively, it is often useful to incorporate batch-normalization layers. These layers standardize the l th channel of an input feature map $h_{\text{in}}[l]$ using its sample mean μ_l and variance σ_l^2 , computed over a training batch (as we discuss in Section V batches are subsets of data used to train the model). The standardized features are then re-centered and re-scaled using the learned parameters γ_l and β_l to yield the output feature map $h_{\text{out}} := \mathcal{B}_{\mathcal{N}} h_{\text{in}}$, whose j th entry equals

$$h_{\text{out}}[l]_j := \gamma_l \cdot \frac{h_{\text{in}}[l]_j - \mu_l}{\sigma_l} + \beta_l. \quad (11)$$

The batch-normalization operation is differentiable with respect to the input feature map and the parameters.

Layers at different depths of a deep neural network can be linked by *skip connections*. For denoising, it is often useful to connect the input to the output, so that the network only needs to learn a residual estimate. This provides the model access to the identity function, which is particularly useful when the available training data is limited. Figure 4 illustrates this for denoising of 1D piecewise-constant functions.

One of the main CNN architectures for image denoising, known as the Denoising CNN (DnCNN) [120], consists of 20 convolutional layers followed by batch normalization layers (except for the first one) and ReLU layers, with a skip connection from beginning to end. The resulting denoising function is therefore of the form

$$\mathcal{D}_{\text{DnCNN}}(y) := y + \mathcal{C}_{20} \mathcal{R}_+ \mathcal{B}_{\mathcal{N}} \mathcal{C}_{19} \cdots \mathcal{R}_+ \mathcal{B}_{\mathcal{N}} \mathcal{C}_2 \mathcal{R}_+ \mathcal{C}_1 y, \quad (12)$$

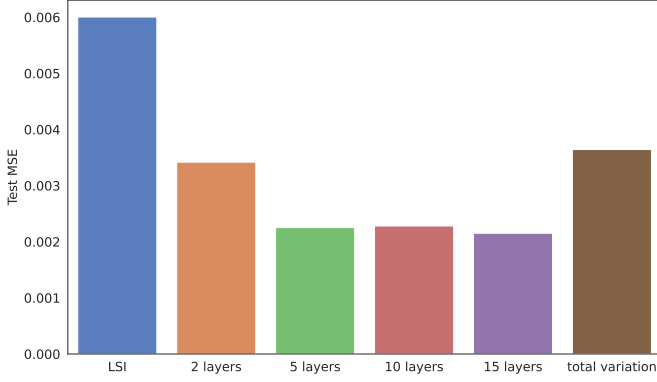


Fig. 3. **CNN denoisers for 1D piecewise constant functions:** The graph compares the test MSE of a LSI filter, total-variation regularization and CNNs with different numbers of layers for denoising 1D piecewise-constant functions corrupted by additive Gaussian noise. All CNNs outperform the linear filter, showing that nonlinearity is important. Additional layers further improve performance, outperforming total-variation regularization, but the improvement is very slight beyond 10 layers. All of the CNNs have skip connections connecting their input to the output.

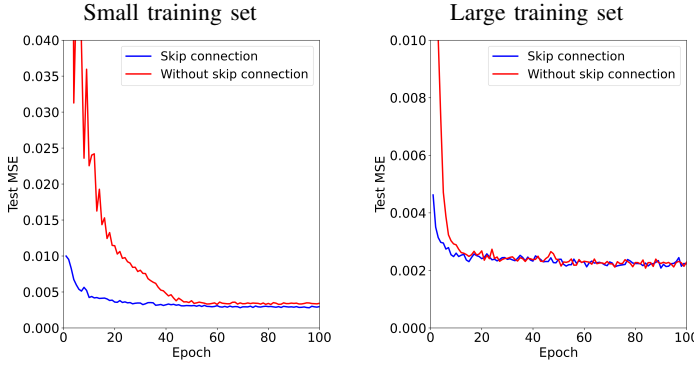


Fig. 4. **Skip connection.** The curves in the graph show the test MSE (as a function of the training epochs) of a 10-layer denoising CNN with (blue) and without (red) a skip connection linking the network input to the output. When the CNN is trained on a small training set with just 10 signals (left) the skip connection is essential; without it training is very unstable. When trained on a larger training set with 500 signals (right), the skip connection still provides faster convergence, but the performance gain is more moderate.

as shown in Figure 5. We use DnCNN for our experiments on natural images.

C. Receptive field and the U-Net

The receptive field of a CNN is the region of the input signal that affects its output. In the case of denoising, it is the set of noisy data that are combined to produce each entry of the denoised signal. A key consideration in the design of a CNN is whether its receptive field is sufficient to capture the input features relevant to the task of interest.

The receptive field depends on the size of the filters in the convolutional layers. Each entry j of the convolution between a signal h and a filter w can be expressed as the inner product between w and a segment or a patch (depending on whether the signal is 1D or 2D) $h_{\{j\}}$ of h centered at j with the same dimensions as the filter,

$$(w * h)_j = \langle w, h_{\{j\}} \rangle. \quad (13)$$

Therefore, the receptive field of a single convolutional layer with respect to its input is equal to the size of the corresponding filter. Assuming that each convolutional layer has filters with the same size (as is the case, for instance, in the DnCNN architecture), the length of the receptive field of a neural network with 1D inputs grows linearly with the number of convolutional layers, whereas the area of the receptive field of a network with 2D inputs grows quadratically. ReLU layers, batch-normalization layers and skip connections do not modify the receptive field.

Building CNNs with large receptive fields requires utilizing very large convolutional filters or many layers. Unfortunately, this can dramatically increase the number of model parameters when the number of channels in each layer is large (as is typically necessary to achieve good performance), which limits the receptive field in practice. For instance, the receptive field of DnCNN is only 41×41 pixels. This is sufficient to achieve accurate denoising for natural images, but can severely hamper performance in scientific-imaging applications. Scientific images typically have a much larger number of pixels (compare the sizes of the images in Figure 5) and denoising them often requires exploiting large-scale features, such as periodic patterns [79]. This is illustrated in Figure 5, which reports the denoising performance of two DnCNN models, one trained on natural images and another on electron-microscopy images. The natural-image DnCNN exhibits strong denoising performance on held-out natural images, but the electron-microscopy DnCNN does not perform well on held-out electron-microscopy images. The reason is that its limited receptive field is unable to leverage the large-scale periodic patterns in the data.

In order to increase the receptive field of a CNN without dramatically increasing the number of parameters, one can use *downsampling* layers, which decrease the dimensionality of their input. This is typically implemented via a *max pooling* operation, where the input feature map is partitioned into 1D segments or 2D patches. Each entry in the output feature map is then set equal to the maximum value in each segment or patch. The receptive field of the downsampling layer is equal to the size of each segment or patch.

The U-Net architecture, originally introduced for the task of image segmentation [92], applies several downsampling layers to obtain a very large receptive field with a modest number of parameters. The intermediate feature maps can be interpreted as a nonlinear multiscale decomposition of the input signal, in the spirit of the multiscale transformations mentioned in Section III. *Upsampling* layers, implemented via transposed convolutions with learned weights, transform these feature maps back to the input dimensionality. The name of the network stems from the presence of skip connections that bypass each downsampling and upsampling level. Figure 5 depicts the different components in the U-Net architecture. U-Nets have been shown to outperform regular CNNs for denoising microscopy images with periodic structures [79, 108]. In our experiments, we use a U-Net to denoise the simulated electron-microscope data, as well as the real data in Section IX. Figure 5 shows that the U-Net has a much large receptive field than DnCNN, which results in superior

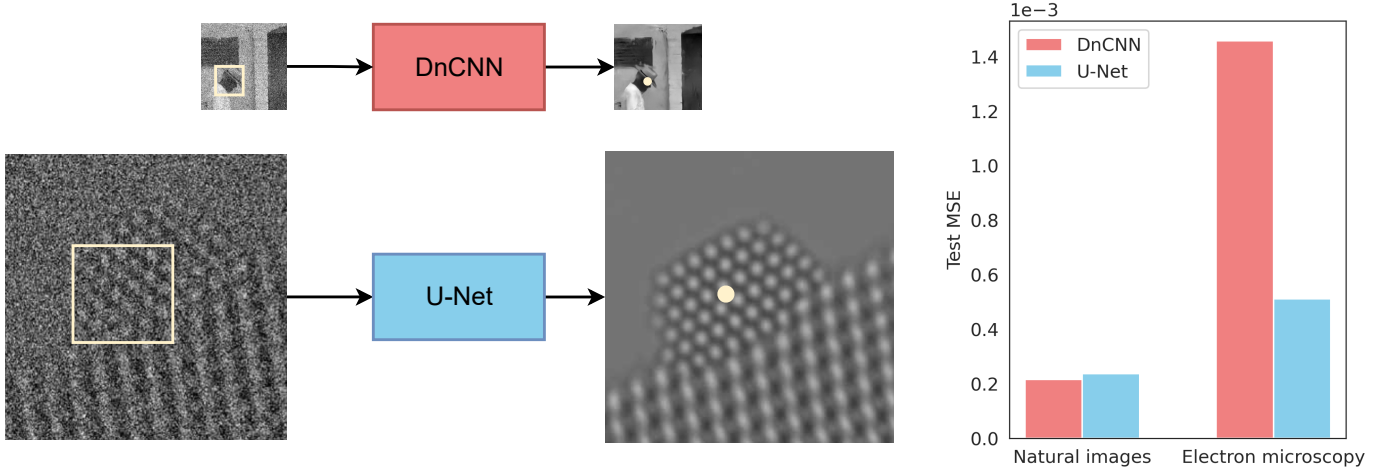


Fig. 5. **Receptive field and denoising performance.** The light-colored square in the top-left diagram on the left depicts the 41×41 receptive field of DnCNN, a popular deep denoiser for natural images. The light-colored square in the bottom-left diagram shows the receptive field of the U-Net model used in our experiments, which is more than 25 times larger (221×221 pixels). The graph on the right compares the denoising performance of a DnCNN and a U-Net trained and tested on natural images (left), and of a DnCNN and a U-Net trained and tested on electron-microscopy images (right). The U-Net clearly outperforms DnCNN on electron-microscopy data, where DnCNN cannot capture large-scale periodic structure due to its small receptive field. In contrast, DnCNN performs slightly better on the smaller-sized natural images, where such a large receptive field is not required for effective denoising. A description of the training and test sets for these experiments is provided in Sections B and C.

performance for the electron-microscopy data.

V. SUPERVISED LEARNING

As explained in Section, CNNs are complicated nonlinear models with a large number of parameters. In this section we explain how to learn these parameters from data in a *supervised* manner, utilizing a *training set* of examples. In the case of denoising, the examples are pairs of noise-free and noisy signals. During training, the parameter values are modified iteratively to minimize the difference between the denoised estimates produced by the CNN and the noise-free signals.

Training neural networks typically requires large datasets, which are divided into subsets called *batches*. The networks are trained by minimizing an appropriate loss or cost function on each batch. In the case of denoising, the loss is the mean squared error (MSE) between the denoised CNN output and the corresponding noise-free example, approximated by the residual sum of squares. Let us denote the CNN by $\mathcal{D}_\Theta$, where Θ is the set of learnable parameters, including convolutional filters, bias parameters, and batch-normalization parameters. The MSE loss over a batch of n_B examples indexed by $\mathcal{B}$ is

$$\text{MSE}_B(\Theta) := \frac{1}{n_B} \sum_{i \in \mathcal{B}} \|x^{[i]} - \mathcal{D}_\Theta(y^{[i]})\|^2, \quad (14)$$

where $x^{[i]}$ is the i th noise-free signal and $y^{[i]}$ the corresponding noisy data. Training is performed by taking a step in the direction opposite to the gradient $\nabla_\Theta \text{MSE}_B(\Theta)$ of the loss with respect to the model parameters,

$$\Theta_b := \Theta_{b-1} - \eta \nabla_\Theta \text{MSE}_B(\Theta). \quad (15)$$

Θ_{b-1} and Θ_b denote the model parameters before and after processing the b th batch. The size of the step is scaled by

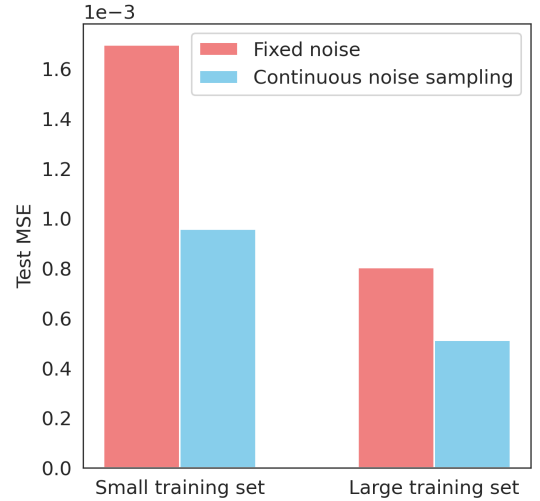


Fig. 6. **Continuous noise sampling.** Comparison of CNN denoisers trained on noisy data with (blue) and without (red) continuous noise sampling, where fresh noise samples are generated at each training epoch. The CNN denoisers are U-Nets trained to denoise electron-microscopy images corrupted with Poisson noise. Continuous noise sampling clearly improves performance, achieving a greater improvement for a small training set with 35 images (left) than for a large training set 350 images (right).

the *learning rate* η , which can depend on b . During training, it is important to tune the value of η correctly. A small value results in very slow progress, whereas a large value can produce instabilities that impede learning. A popular strategy is to utilize methods such as Adam [55], which adapt the learning rate based on estimates of lower-order moments of the gradient. Here, we use Adam in all of our computational experiments.

The batch gradient in (15) is obtained via *backpropagation*, an efficient implementation of the chain rule that propagates

derivatives backwards through the network (see Section 6.5 in [40]). This learning procedure is known as *stochastic gradient descent*, because the batches are usually obtained by sampling randomly from the training data, so that the batch loss MSE_B can be interpreted as a stochastic approximation of the MSE computed over the whole training set.

Performing gradient steps over all of the batches in the training set completes an *epoch*. At each epoch, it is advisable to compute a training and validation loss, which quantify the fit of the model to the training set and to a held-out validation set (separate from the training set) respectively. A model with a large training error is said to *underfit* the training set. A model with a small training error but a large validation error is said to *overfit*. In order to avoid overfitting as much as possible, the model parameters are typically selected to correspond to the epoch with minimum validation loss.

Figure 1 reports the performance of CNNs trained to denoise three types of data: 1D piecewise-constant functions with additive Gaussian noise (left column), natural images with additive Gaussian noise of varying variance (middle column) and electron-microscopy data with Poisson noise (right column). Detailed descriptions of the datasets, CNN models and training procedure are provided in Appendices A, B, C, D and E. The CNNs are compared to a linear shift-invariant filter, which approximates the Wiener filter based on signal statistics from the training sets, and to nonlinear baselines, chosen for their strong performance in each setting. These nonlinear baselines are total-variation regularization [14] for the 1D piecewise constant functions, BM3D [22] for the natural images, and PURE-LET [69] for the electron-microscopy data. The test metrics achieved by each denoiser in a held-out test set are reported below each example. The metrics are PSNR for all signals (see Section II), the perceptual metric SSIM for the two type of images [111], and the perceptual metric LPIPS for the natural images [121].

In our experiments, the supervised CNN outperforms the other methods in all metrics, although BM3D achieves similar results for natural images. Note, however, that the electron-microscopy image in Figure 1 denoised by the supervised CNN (third row, fifth column) contains a small spurious atom at the edge of the nanoparticle, which is not present in the noise-free ground truth (third row, first column). This illustrates an important pitfall of deep denoisers: they may *hallucinate* signal features present in the training data, which are consistent with the observed noise, but not with the true underlying signal. This is particularly problematic for applications to scientific discovery, and is not necessarily captured by traditional evaluation metrics, as discussed in Section VIII.

The supervised-learning framework assumes the availability of a set of noise-free and noisy signal pairs. In practice, obtaining such pairs requires acquiring data simultaneously at a very high SNR to obtain the noise-free signal, and a lower SNR to obtain its noisy counterpart. However, in many applications it may not be possible to perform simultaneous acquisitions, or to measure data at sufficiently high SNR. In such cases, if an accurate model for the data-acquisition process is available, the noise-free-noisy pairs can be gener-

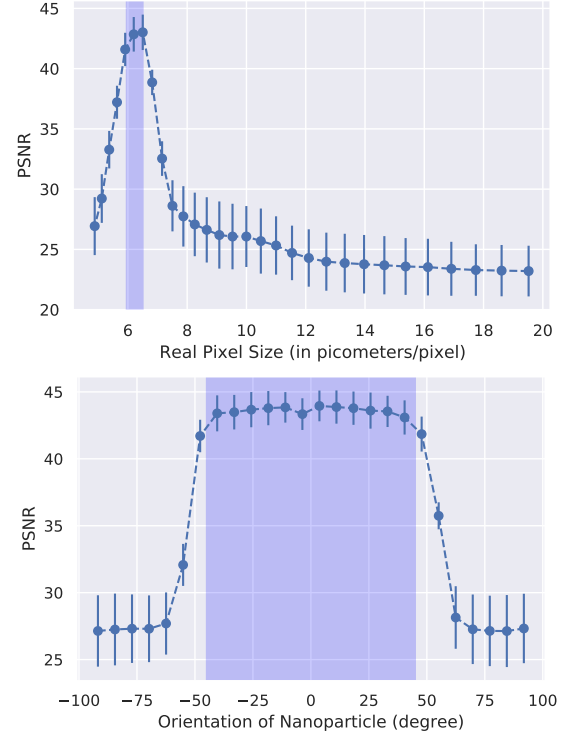


Fig. 7. **Overfitting scaling and orientation.** The plots show results from computational experiments reported in [79], illustrating how CNNs overfit the characteristics of the training data. In the top graph a U-Net is tested on simulated images with different scalings, and therefore different real pixel sizes (measured in picometers). Performance is markedly better for the shaded values, which are included in the training set. Similarly, in the bottom graph the model is tested on images with different orientations (achieved by rotating reference images where a nanoparticle is on top of a support material). Performance again degrades substantially for orientations not present in the training data (indicated by the shaded values).

ated via computational simulations: first simulating the noise-free signal and then corrupting it with synthetic noise [79]. When using simulated data, it is often advantageous to perform *continuous noise sampling*, generating new noisy data from the noise-free signals at each epoch. This strategy can significantly boost performance when the number of training data is limited, as reported in Figure 6, because it prevents overfitting to a fixed set of noise realizations.

A common pitfall when deploying deep-learning models in practice is that they may not *generalize* when their input is different to the data used for training, a phenomenon known as *distribution shift*. For example, if we train a denoising CNN on an electron-microscopy dataset with a certain pixel resolution or a fixed orientation, it may underperform when applied to images with different resolutions and orientations. Figure 7 shows the results of an experiment reported in [79], which demonstrates that the decrease in performance can be very significant. It is therefore crucial to ensure that the training data are representative of the data encountered when the model is deployed, as discussed further in Section IX-B.

VI. WHAT DO DEEP DENOISERS LEARN?

Deep neural networks implement highly complex nonlinear functions, which are notoriously challenging to interpret. In

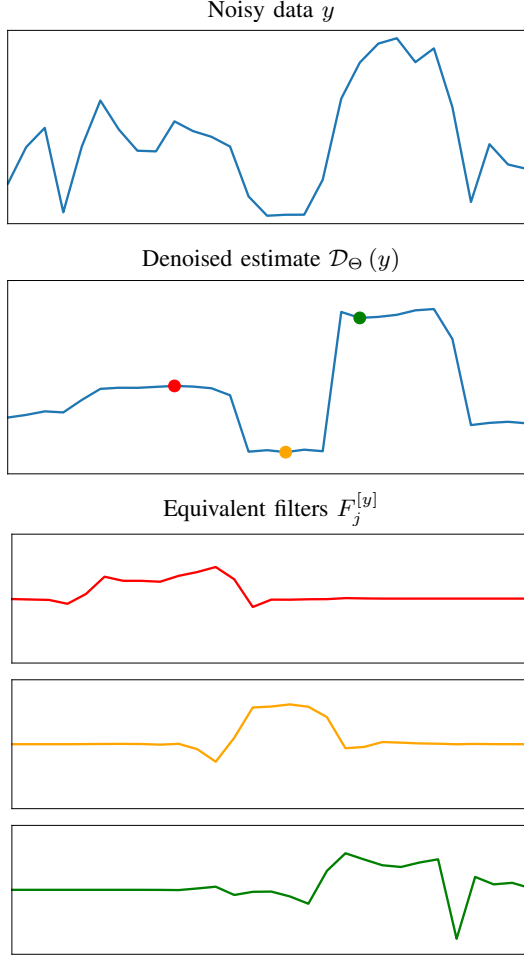


Fig. 8. **Denoising CNNs learn to perform adaptive averaging.** The top image shows a 1D piecewise-constant signal y corrupted by additive Gaussian noise. The image below shows the result of applying a denoising CNN (described in Appendix A) to the noisy data $\mathcal{D}_\Theta(y)$. The three bottom images show equivalent filters $F_j^{[y]}$, equal to the partial derivatives of the corresponding colored entry j of the denoised estimate with respect to the noisy input, as defined in Section VI. The equivalent filters reveal that the CNN learns to perform approximate adaptive averages within the piecewise-constant segments of the signal, without blurring the segment boundaries.

this section, we describe a simple analysis technique that provides insight into the strategies learned by deep CNNs trained for denoising. The approach follows [77], which proposes to interpret nonlinear denoisers by decoupling their implemented function into a first phase, in which highly nonlinear decisions are made, and a second phase, in which such decisions result in an adaptive linear filter that is applied to the noisy data. In the case of denoising CNNs, these adaptive filters are encoded in the derivatives of the deep denoiser with respect to its input [80]. Visualizing such derivatives is a popular strategy to interpret deep-learning classifiers [98, 81].

For any fixed input $y \in \mathbb{R}^d$, we can express the function implemented by a deep convolutional neural network $\mathcal{D}_\Theta$, like the ones described in Section IV, as an affine function of y . Convolutional layers, batch-normalization layers and skip connections all implement affine functions. In addition, when the input to the network is fixed, the ReLU layers are equivalent to pointwise multiplications by a binary mask that

encodes which entries are preserved, and which are set to zero. Consequently, we can represent the denoising function as a cascade of affine functions of the input, which is itself an affine function of the input. More precisely, there exists a matrix $F^{[y]} \in \mathbb{R}^{d \times d}$ and a vector $b^{[y]} \in \mathbb{R}^d$, such that

$$\mathcal{D}_\Theta(y) = F^{[y]}y + b^{[y]}. \quad (16)$$

The superscript on $F^{[y]}$ and $b^{[y]}$ highlight that both depend on the ReLU activation patterns, and therefore on the input vector y . The vector $b^{[y]}$ is the *net bias* of the network for that input. It is a combined function of all additive constants in the CNN, after propagating them through the network. It has been shown that suppressing the net bias, by removing all additive constants in the network, improves the generalization of denoising CNNs to different levels of Gaussian noise [80].

The matrix $F^{[y]}$ is equal to the Jacobian matrix $\nabla_y \mathcal{D}_\Theta(y)$ of the CNN with respect to its input. This is a $d \times d$ matrix, which contains the partial derivatives of each entry of the denoised signal with respect to each entry of the noisy input, *not* the derivatives with respect to the network parameters, as computed during training (see Section V).

Equation (16) makes it possible to express the j th denoised entry of the output $\mathcal{D}_\Theta(y)_j$ as an inner product between the noisy input y and the j th row $F_j^{[y]}$ of the Jacobian matrix added to the j entry $b_j^{[y]}$ of the net bias:

$$\mathcal{D}_\Theta(y)_j = \langle F_j^{[y]}, y \rangle + b_j^{[y]}. \quad (17)$$

We can therefore interpret $F_j^{[y]}$ as an *equivalent filter* that is applied to the noisy input y in order to obtain the j th entry of the denoised signal. In order to interpret the denoising function learned by the CNN, we visualize the equivalent filters corresponding to different entries. These filters can be obtained efficiently using automatic differentiation in packages like PyTorch or TensorFlow. As a word of caution, the affine approximation (17) provides a *local* description of the denoising function. Modifications of the input that change the activation pattern of the CNN, will completely alter the denoising function, invalidating the affine approximation. This is not surprising; their ability to adapt nonlinearly to different inputs is precisely one of the main advantages of CNNs.

Figure 8 depicts equivalent filters corresponding to three entries in a 1D piecewise-constant function. The filters represent an approximate average of the noisy entries corresponding to constant segments, while respecting the boundaries between the segments. This is reminiscent of the nonlinear-filtering approaches mentioned in Section III, which average neighboring pixels with similar intensities. The CNN is able to learn such strategies automatically from the training data.

Figure 9 shows the equivalent filters corresponding to two pixels in a natural image, when the input to a CNN is corrupted by two different noise levels. The CNN is trained with different noise levels, as described in Appendix B. As in the case of the 1D piecewise-constant function, the equivalent filters represent an approximate average over the surrounding noisy pixels, which adapts to signal structure in the spirit of nonlinear filters like the bilateral filter. Interestingly, the averaging operation

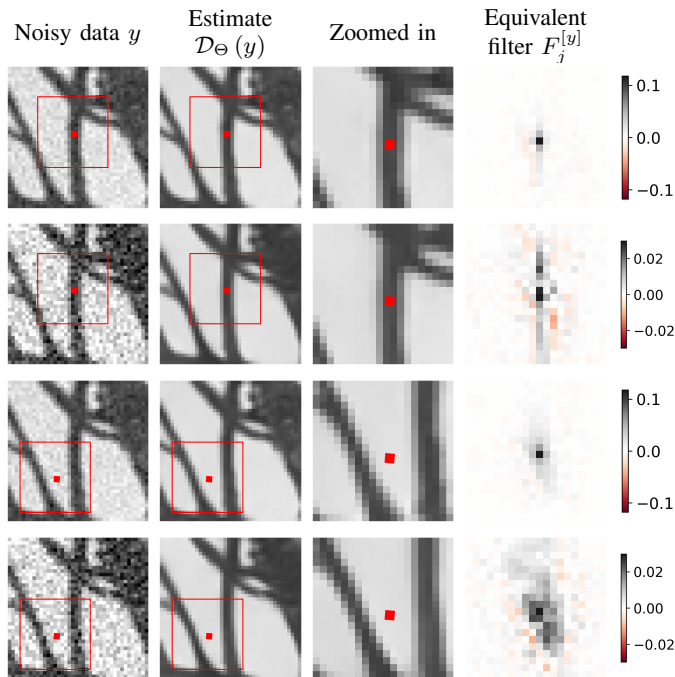


Fig. 9. **Denoising CNNs adapt to signal structure and noise level.** The first column shows part of a natural image y corrupted by additive Gaussian noise. The second and third columns show the denoised output $\mathcal{D}_\Theta(y)$ produced by a CNN trained on varying noise-levels. The fourth column shows the equivalent filter $F_j^{[y]}$, defined in Section VI, which equals the partial derivative of an output pixel j (indicated by a red dot in the three first columns) with respect to the network input. The equivalent filters reveal that the network implements an adaptive average over regions where the intensity of the underlying noise-free image is approximately constant. The average adapts to the noise statistics of the input image, encompassing a larger region when the noise level increases (compare the second row with the first, or the fourth with the third), but without blurring object boundaries. The CNN is the bias-free version of DnCNN described in Appendix B.

adapts to the noise level, encompassing more pixels when the noise increases (compare the second row to the first, and the fourth to the third). As this occurs, the filters remain within object boundaries, avoiding the edge blurring that is commonly observed in linear shift-invariant filtering.

Figure 10 depicts equivalent filters corresponding to noisy pixels in the real electron-microscopy data analyzed in Section IX. The filters represent an approximate local average that adapts to the periodic atomic pattern in the signal. This shows that the network learns how to exploit localized periodicities, which is notoriously challenging and one of the motivations for the development of wavelets [72].

VII. UNSUPERVISED LEARNING

A. Motivation and intuition

The supervised-learning framework described in Section V requires a database of ground-truth noise-free examples, along with corresponding noisy data. In scientific applications building such databases can be very challenging, or even impossible when the goal is to uncover unobserved phenomena, since they cannot be simulated. This motivates the need for unsupervised methods, capable of learning directly from noisy data.

In this section we describe several approaches that enable the training of deep denoisers using only noisy data. We

focus on approaches that can adapt to general, and potentially unknown, noise distributions. An alternative approach, tailored to additive Gaussian noise, is to utilize Stein’s unbiased risk estimator (SURE) as a training cost function [76, 100, 126, 78]. This approach provides an asymptotically unbiased estimator of the MSE under the Gaussian assumption [26], which was leveraged by several denoising methods before the deep-learning era [88, 68].

In supervised learning, the model parameters Θ are learned by minimizing the MSE between the noise-free ground-truth signals $x^{[i]}$ and the model estimate $\mathcal{D}_\Theta(y^{[i]})$ ($1 \leq i \leq n$),

$$\text{MSE}(\Theta) := \frac{1}{n} \sum_{i=1}^n \left\| x^{[i]} - \mathcal{D}_\Theta(y^{[i]}) \right\|^2. \quad (18)$$

Unsupervised learning seeks to achieve a similar effect, without requiring access to the noise-free signals.

As a motivating thought experiment, imagine replacing the noise-free signal $x^{[i]}$ by the corresponding noisy measurement $y^{[i]}$, acting as a *target* that is a proxy for the noise-free signal. The resulting naive unsupervised loss equals

$$\text{Naive}(\Theta) := \frac{1}{n} \sum_{i=1}^n \left\| y^{[i]} - \mathcal{D}_\Theta(y^{[i]}) \right\|^2. \quad (19)$$

The loss is naive, because it can be trivially minimized by a useless denoiser that implements the identity function and maps its input directly to its output. However, perhaps surprisingly, one can obtain a non-trivial denoiser with this naive loss, even with a single example! This remarkable result was achieved in [107] by constraining the model complexity via a bottleneck layer with low dimensionality and applying early stopping. Intuitively, the model is able to learn image structure first, before eventually overfitting the noise. Nonetheless, this approach is fundamentally limited, as it is very challenging to avoid overfitting when leveraging models with the high complexity required for effective denoising.

In Sections VII-B and VII-C we present two unsupervised frameworks to train deep denoisers, which build upon the same principle of directly estimating noisy data as the naive unsupervised loss above, combined with two different strategies to avoid overfitting. The two frameworks, along with the naive loss, are illustrated in Figure 11.

B. The Noise2Noise framework

The key issue with our naive unsupervised loss is the dependence between the noise in the target and the model output. To overcome this, the Noise2Noise [64] method uses a target that is a noisy proxy of the noise-free signal, but also *independent from the input noise*. Intuitively, approximating such a target can only be achieved by estimating the underlying noise-free signal, since the input noise cannot be estimated from it. To justify the approach mathematically, let us focus on the j th entry x_j of a noise-free signal x . We model the corresponding observed noisy entry as the random variable $\tilde{y}_j$, centered at x_j so that $\mathbb{E}[\tilde{y}_j] = x_j$. Assume that we have access to a noisy target modeled as the random variable $\tilde{t}_j$, also centered at x_j ($\mathbb{E}[\tilde{t}_j] = x_j$), but independent from $\tilde{y}_j$. Then,

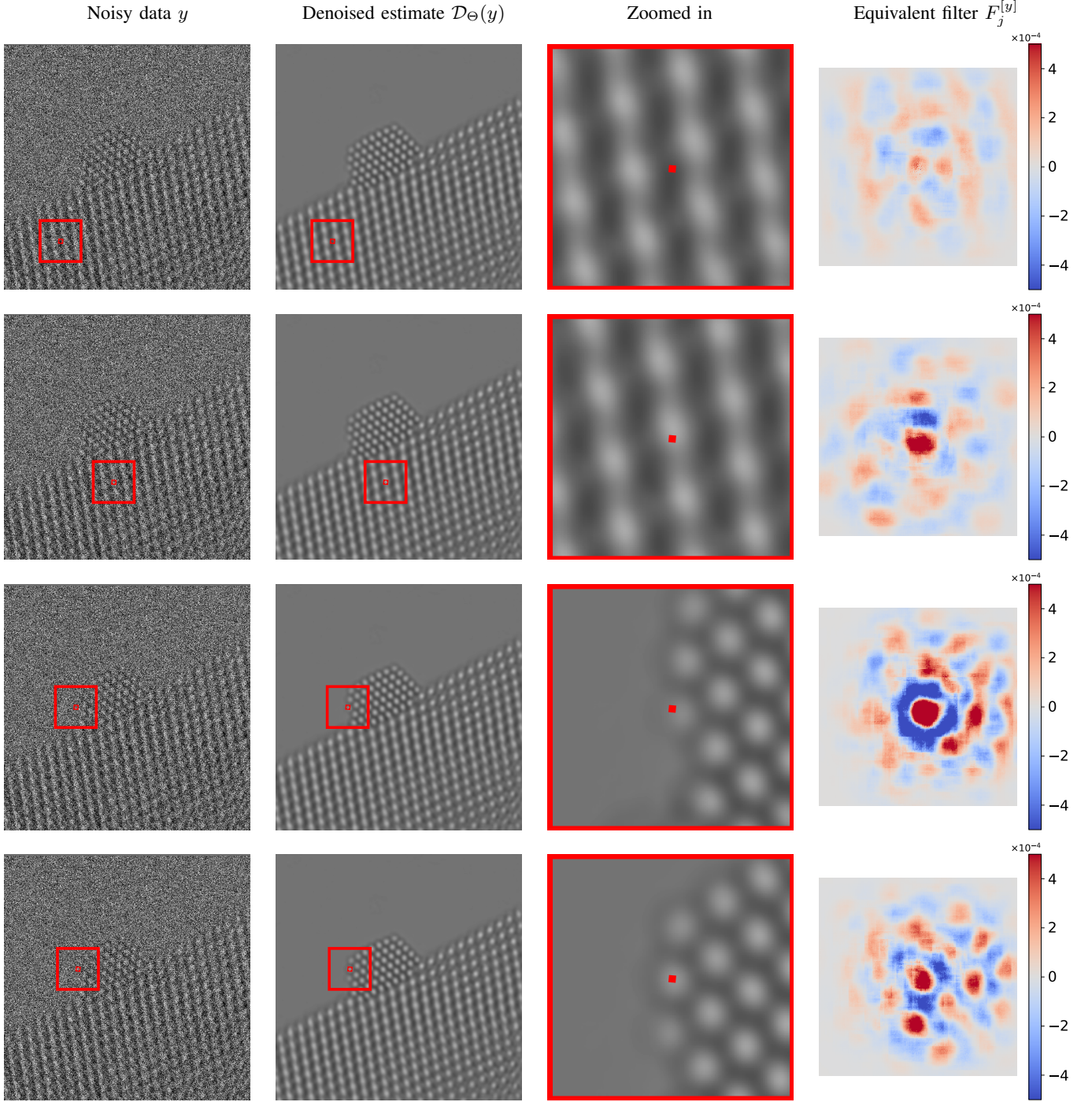


Fig. 10. **Denoising CNNs jointly exploit local and periodic structure.** The first column shows a frame of real-world transmission-electron-microscopy noisy data y . The second and third columns show the denoised output $\mathcal{D}_{\Theta}(y)$ produced by a CNN. The fourth column shows an equivalent filter $F_j^{[y]}$, containing the partial derivatives of an output pixel j (indicated by a red dot in the first three columns) with respect to the network input as described in Section VI. The equivalent filters reveal that the CNN is able to exploit local structure via averaging, while taking into account the underlying periodic structure. The CNN is a U-Net trained on a restricted set of simulated images, as explained in Section IX-B.

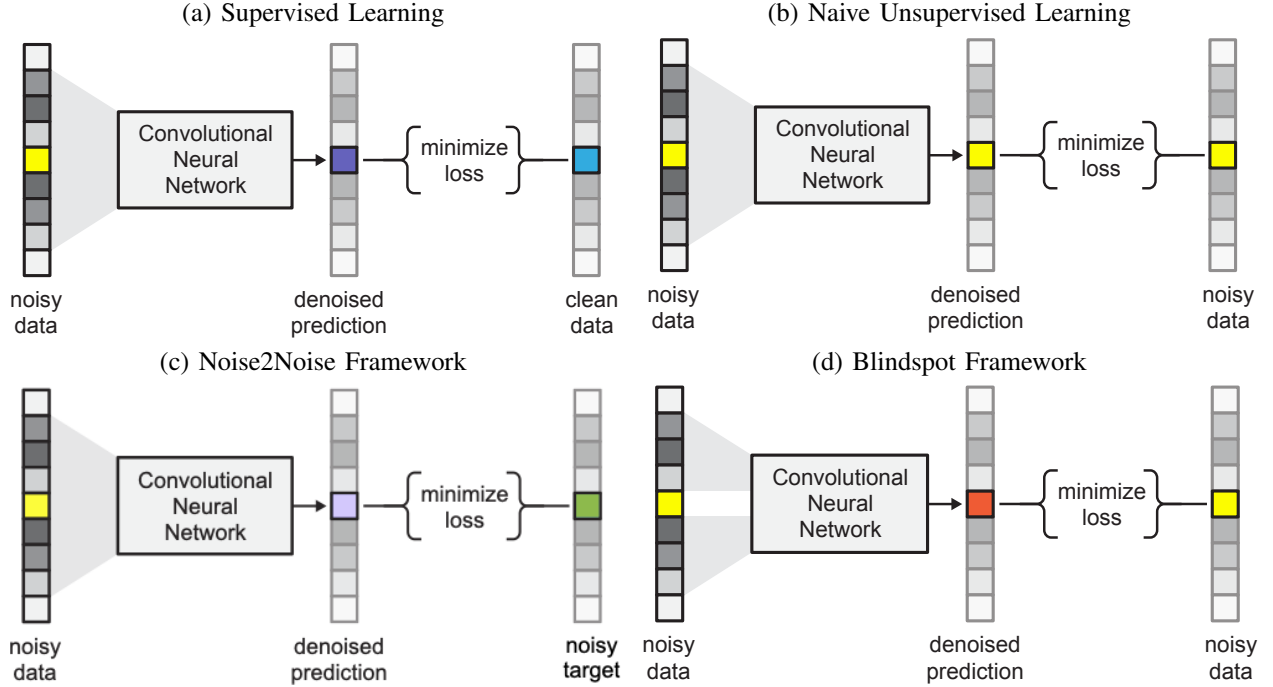


Fig. 11. **Supervised and unsupervised training of deep denoisers.** (a) Supervised learning trains CNNs to approximate a ground-truth noise-free signal from the available noisy data. (b) Replacing the noise-free signal by the noisy data yields a naive unsupervised loss that can be minimized by setting the denoised estimate equal to the noisy input. (c) The Noise2Noise framework seeks to instead approximate a noisy target that corresponds to the same underlying noise-free signal as the noisy input, but is corrupted by independent noise. Consequently, minimizing the resulting error can only be achieved by estimating the noise-free signal. (d) The blindspot framework seeks to approximate each noisy entry, from a modified input that ignores that specific entry. If the noise is entry-wise independent, this again can only be achieved by estimating the noise-free signal.

the MSE between the target and the corresponding denoised entry $\mathcal{D}_\Theta(\tilde{y})_j$ equals

$$\begin{aligned} & \mathbb{E} \left[\left(\tilde{t}_j - \mathcal{D}_\Theta(\tilde{y})_j \right)^2 \right] \\ &= \mathbb{E} \left[\left(\tilde{t}_j - x_j \right)^2 \right] + \mathbb{E} \left[\left(x_j - \mathcal{D}_\Theta(\tilde{y})_j \right)^2 \right], \end{aligned} \quad (20)$$

as derived in Section H. Summing this expression over all the entries in the signal, we conclude that minimizing the ℓ_2 -norm error between the target and the denoised signal to estimate the model parameters is *equivalent to minimizing the supervised MSE*:

$$\begin{aligned} & \arg \min_{\Theta} \mathbb{E} \left[\left\| \tilde{t} - \mathcal{D}_\Theta(\tilde{y}) \right\|_2^2 \right] \\ &= \arg \min_{\Theta} \mathbb{E} \left[\left\| x - \mathcal{D}_\Theta(\tilde{y}) \right\|_2^2 \right]. \end{aligned} \quad (21)$$

Noise2Noise leverages this insight to train denoising CNNs in a *fully unsupervised manner*, assuming the availability of a target $t^{[i]}$ associated with each example ($1 \leq i \leq n$) that satisfies two properties: (1) It corresponds to the same noise-free signal as the noisy data $y^{[i]}$. (2) It is corrupted by noise that is independent from the noise in $y^{[i]}$. The resulting unsupervised training loss equals:

$$\text{Noise2Noise}(\Theta) := \frac{1}{n} \sum_{i=1}^n \left\| t^{[i]} - \mathcal{D}_\Theta(y^{[i]}) \right\|^2. \quad (22)$$

In practice, it may be challenging to obtain targets that satisfy the requirements of the unsupervised loss, even approximately. A possible strategy is to perform several signal measurements within a short time interval. However, the resulting data only satisfies the conditions if the underlying noise-free signal remains fixed (and the noise is independent across acquisitions). Several methods extend this approach to denoise videos of dynamic scenes by registering the frames using an estimate of the optical flow [28, 118, 44, 24].

The Neighbor2Neighbor method [49] constructs the target-input pairs in (22) from individual noisy signals $z^{[1]}, \dots, z^{[n]}$. Each signal $z^{[i]}$ is spatially subsampled to form the target $t^{[i]}$ and the noisy input $y^{[i]}$. Subsampling should be carried out so that the target and input are as similar as possible, but without sharing any entries. For this approach to succeed, the underlying noise-free signal must be sufficiently smooth, to ensure that neighboring entries are almost equal, and the noise must be entry-wise independent. Under these two assumptions, the target and input are approximately equal to the same ground-truth noise-free signal with independent noise, approximately satisfying the requirements for equivalence between the unsupervised loss (22) and the supervised MSE.

Figure 1 reports the performance of CNNs trained following the Noise2Noise framework in an idealized setting, where the target corresponds to the same noise-free signal with independent noise, for our three example datasets. For 1D piecewise-constant functions (left column) and natural images (middle column) the unsupervised models achieve similar performance to the supervised CNNs. For the electron-microscopy data,

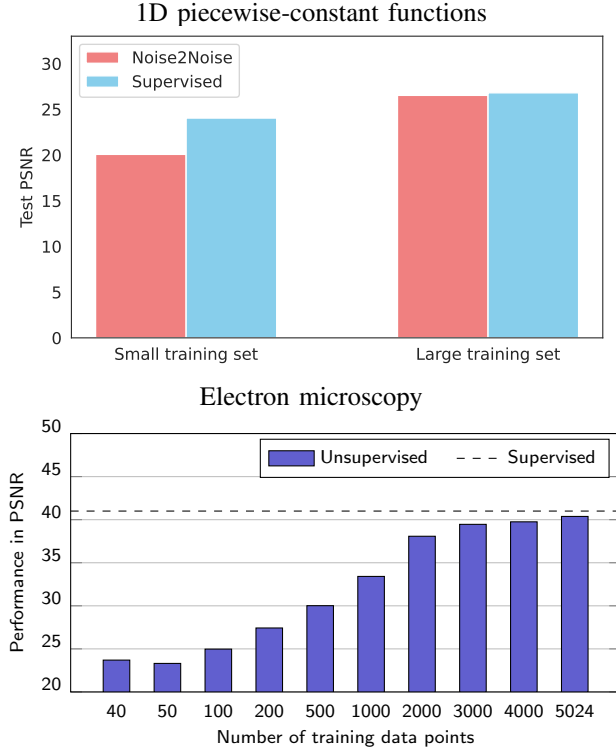


Fig. 12. **Effect of training set size on unsupervised denoising.** The top graph compares the test performance of CNNs trained following the Noise2Noise framework described in Section VII-B and the supervised-learning framework described in Section V. The CNNs are trained using two different-sized datasets of 1D piecewise-constant signals. Performance is noticeably worse for the small training set with 10 signals, but essentially the same as the supervised model for the larger training set with 500 signals. The bottom plot shows the result of an experiment reported in [79], where a blindspot network is trained on training sets of different sizes (containing simulated electron-microscopy images). The performance is again poor when the training set is small, but improves to the level of supervised approaches (denoted by dashed line) when utilizing more training data.

the performance is better than the LSI filter, but 3 dB worse than the supervised model. The reason is the limited number of training data. Unsupervised methods tend to require large training sets to emulate the performance of supervised approaches, as illustrated in Figure 12. However, note the unsupervised denoiser does not hallucinate a spurious atom for the electron-microscopy image shown in Figure 1 (third row, compare the first and sixth column), unlike the supervised denoiser (third row, fifth column), which would make the result preferable for the purpose of scientific discovery.

C. Blindspot methods

An alternative approach to obtain an unsupervised denoising loss is to *blind* the network, so that it estimates the j th entry from its surrounding spatial neighborhood but *without including the entry itself*, as illustrated in Figure 11. If the noise is independent across entries, then this ensures that the network cannot overfit the noise, and must instead learn to estimate the noise-free component in the noisy data.

Let $\tilde{y}$ be a random vector representing the noisy data corresponding to a noise-free signal x , and let $\tilde{y}_{-j}$ denote the signal obtained by removing the j th entry $\tilde{y}_j$ from $\tilde{y}$. We

consider a *blindspot* CNN $\mathcal{B}_\Theta$ where the j th entry of the output does not depend on the j th entry of the input (but does depend on the other entries), so that

$$\mathcal{B}_\Theta(\tilde{y})_j = \mathcal{B}_\Theta(\tilde{y}_{-j})_j. \quad (23)$$

Assuming that $\tilde{y}_j$ is centered at the j th noise-free entry x_j (i.e. $\mathbb{E}[\tilde{y}_j] = x_j$), and the noise is entry-wise independent, so that $\tilde{y}_j$ and $\mathcal{B}_\Theta(\tilde{y})_j$ are independent (because $\tilde{y}_j$ and $\tilde{y}_{-j}$ are independent), then by the same argument used to derive (20),

$$\begin{aligned} & \mathbb{E} \left[\left(\tilde{y}_j - \mathcal{B}_\Theta(\tilde{y})_j \right)^2 \right] \\ &= \mathbb{E} \left[\left(\tilde{y}_j - x_j \right)^2 \right] + \mathbb{E} \left[\left(x_j - \mathcal{B}_\Theta(\tilde{y})_j \right)^2 \right]. \end{aligned} \quad (24)$$

Summing over all the entries, this implies that fitting the noisy data with the blindspot network is equivalent to minimizing the supervised MSE:

$$\begin{aligned} & \arg \min_{\Theta} \mathbb{E} \left[\|\tilde{y} - \mathcal{B}_\Theta(\tilde{y})\|_2^2 \right] \\ &= \arg \min_{\Theta} \mathbb{E} \left[\|x - \mathcal{B}_\Theta(\tilde{y})\|_2^2 \right]. \end{aligned} \quad (25)$$

The resulting unsupervised training loss equals:

$$\text{Blindspot}(\Theta) := \frac{1}{n} \sum_{i=1}^n \left\| y^{[i]} - \mathcal{B}_\Theta(y^{[i]}) \right\|_2^2. \quad (26)$$

Originally, the blindspot approach was implemented as the Noise2Void method, where pixels are masked by replacing them with random values [58] (see also [8] for a related approach known as Noise2Self, and [59, 87, 54] for different extensions). Subsequently, a specialized CNN architecture was proposed, which is designed to explicitly enforce a blindspot in the receptive field [60], a strategy that has also been leveraged for unsupervised video denoising [95].

A disadvantage of the blindspot approach is that the denoising model estimates each signal entry without taking into account the corresponding noisy entry (which is the most informative entry in the noisy data!). If the distribution of the noise is known, one can perform a weighted average of the blindspot-denoised entry and the noisy measurement (see [60, 59] for a derivation assuming additive Gaussian noise). Otherwise, the measurement must be ignored, which can be problematic if the underlying noise-free signal is not sufficiently smooth.

VIII. EVALUATION

In the literature, deep denoising methods are typically evaluated on benchmark datasets where noise-free signals are corrupted with synthetic noise [120, 122, 64, 8]. The evaluation metric is often the MSE or PSNR between the denoised and noise-free signals computed on a held-out set, separate from the training data.

An important challenge in scientific-imaging applications is how to evaluate denoising methods when noise-free signals are not available. Existing benchmark datasets based on real noisy data [1, 86, 115, 54] utilize reference images, obtained by averaging multiple noisy images, as the *noise-free* signals. Unfortunately, averaging suppresses dynamically-changing features, which are of interest in many applied

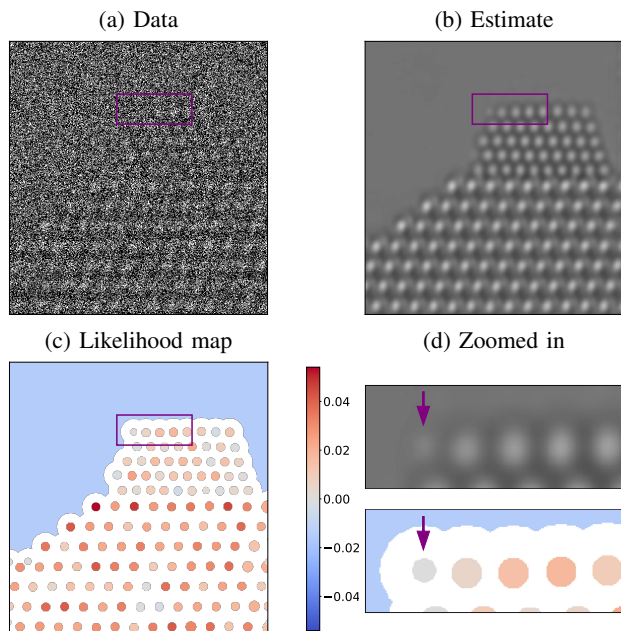


Fig. 13. **Likelihood map.** The figure shows an example of the likelihood-map technique from [79]. When the simulated noisy data in (a) are denoised by a CNN (b), a spurious atom appears at the left edge of the nanoparticle in the denoised estimate (see zoomed-in image (d)). Panel (c) shows the likelihood map, which equals the difference of the log-likelihood of the data assuming the presence and absence of the atom, as explained in Appendix I. The value of the likelihood map at the location of the spurious atom is low (d), which indicates that the presence of an atom is less consistent with the observed data than its absence.

domains. Alternatively, domain scientists may rely on visual inspection to assess denoising on real data, but this only provides a qualitative evaluation.

The unsupervised losses described in Section VII do not provide an accurate estimate of denoising performance. Equations (20) and (24) show that they can be expressed as the sum of the supervised MSE and an additional term. As a result, they can be utilized for training, and for comparing different models on a fixed dataset, but not as a standalone denoising metric. However, recent work shows that it is possible to correct for the additional term in (20) utilizing additional independent noisy targets [73]. The resulting metric is called the unsupervised MSE (uMSE), which can be used to compute an unsupervised PSNR (uPSNR) following (3). The uMSE and uPSNR are unbiased and consistent estimators of the supervised MSE and PSNR [73].

As mentioned in Section II, the MSE and PSNR are not holistic measures of signal quality. In the case of natural images, it is therefore advisable to complement them with metrics designed to evaluate perceptual quality [124, 111, 121]. In other imaging applications, the key question is whether the denoised signals contain the correct scientific information of interest. For example, if the goal is to identify the atomic structure of a material, as in our electron-microscopy example, then one should evaluate whether individual atoms are detected accurately [79]. In that case, the result produced by supervised learning in Figure 1 for the simulated electron-microscopy data could be problematic, as a small spurious atom is present at the edge of the nanoparticle.

Unfortunately, detection accuracy can only be evaluated using simulated data where the ground-truth atomic structure is known. On real data, an alternative is to use a *likelihood map*, a technique introduced in [79] to quantify the agreement between the denoised atomic structure and the observed data. As explained in more detail in Appendix I, the likelihood map is equal to the difference in the log-likelihood of different hypotheses about the signal of interest, based on the denoised estimate. Figure 13 shows an example likelihood map from [79], where the hypotheses correspond to the presence or absence of atoms in a nanoparticle. In the figure, a spurious atom is detected on the left of the zoomed region. The likelihood map at that location is very low, indicating that the presence of an atom is not consistent with the observed data.

IX. A REAL-WORLD APPLICATION OF DEEP DENOISING

Most of the methods described in Sections V and VII have been predominantly applied to images corrupted with synthetic noise. This simplifies quantitative evaluation, but also ignores critical challenges that arise in scientific applications. In this section we illustrate the application of deep denoising to a real-world problem: investigating the behavior of catalytic nanoparticles from noisy transmission-electron-microscopy data.

We show that deep neural networks are able to achieve impressive denoising performance on these real data, and explain the key insights that make this possible. As described in Section IX-B, a supervised approach following Section V requires careful design of the training set to ensure accurate denoising. Section IX-C demonstrates that the unsupervised framework from Section VII can be applied successfully without any training data (as long as held-out noisy validation data is used to prevent overfitting), but underperforms due to the limited quantity of available data. Finally, Section IX-D shows that the supervised and unsupervised approaches can be combined to achieve strong performance, without needing to carefully curate the training data.

A. Application: Electron microscopy imaging of nanoparticles

We consider a noisy dataset depicting a catalytic system at a resolution that enables the identification of individual atoms. The catalytic system consists of platinum nanoparticles lying on a CeO_2 substrate. This bi-functional system, and similar metallic nanoparticles supported on different oxides, are widely used in noise-free energy conversion and environmental remediation applications, in addition to a broad range of other chemical reactions [82, 119, 83]. The data were captured using a transmission electron microscope under the experimental conditions described in Appendix F, and is publicly available.

The data were originally acquired to study the dynamic behavior of the nanoparticles, with the ultimate goal of characterizing catalytic-relevant configurations. This requires imaging at very high frame rates under reaction conditions, which results in severe noise degradation. As a result, the dataset has very different properties to the images in most denoising benchmarks. For instance, the SNR is approximately

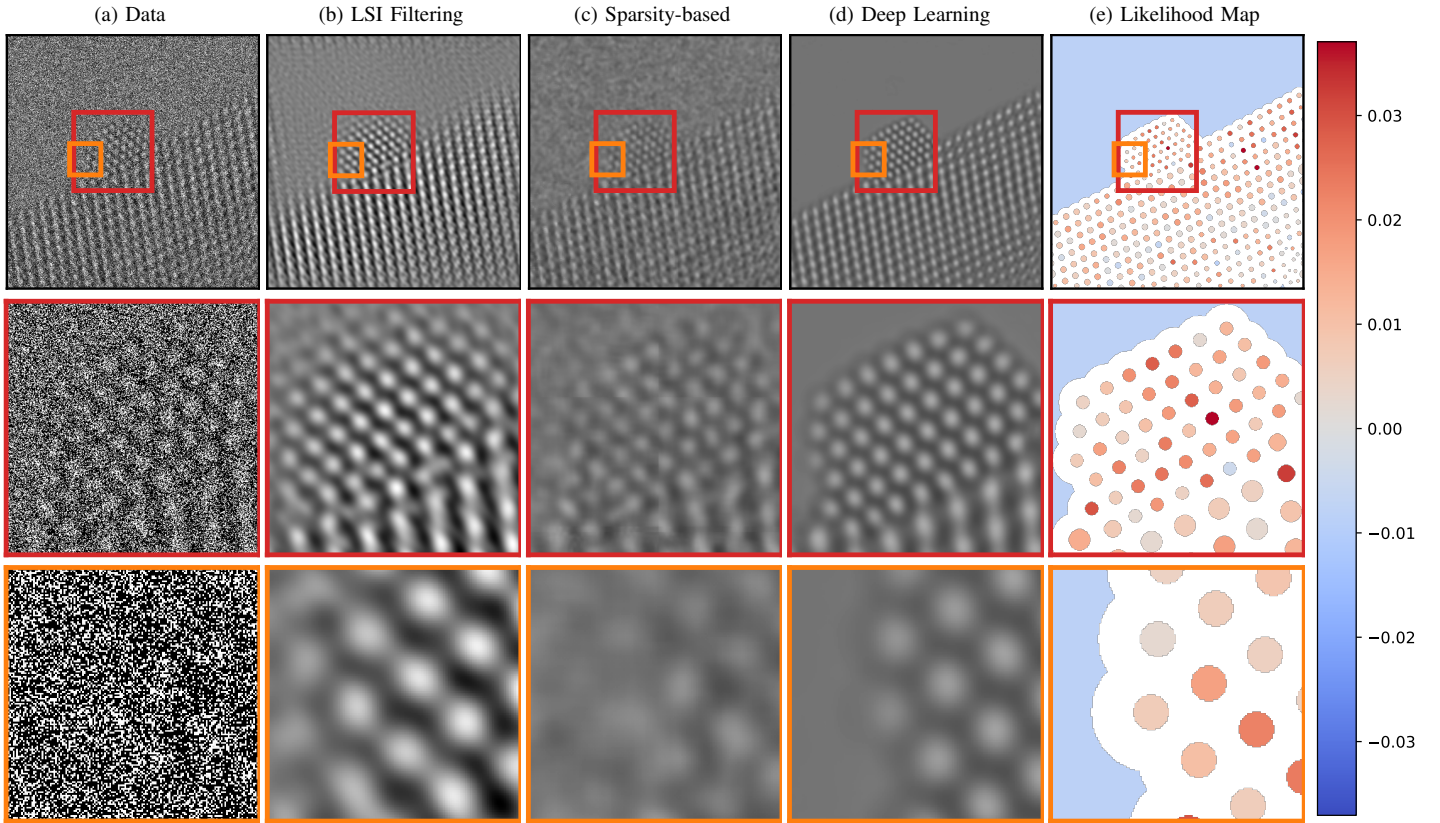


Fig. 14. **Comparison of deep denoising and traditional methods on real data.** Results from [79] comparing the performance of different denoising approaches on real transmission-electron-microscopy data with very low SNR (a), acquired as explained in Appendix F. The data depict a platinum nanoparticle lying on CeO_2 substrate. Linear shift-invariant filtering (b) and a sparsity-based method [69] (c) are clearly outperformed by a deep CNN trained on simulated data (d), which is able to identify the atomic structure of the nanoparticle and the substrate. The likelihood map (see Section VIII) (e) quantifies to what extent the recovered structure is consistent with the data.

3 dB, while most works consider photographic images with an SNR above 22 dB (see e.g. [120]).

Figure 14 shows the result of applying several methods to the dataset (see [79] for a detailed description of implementation details). Traditional approaches based on LSI filtering and sparse estimation struggle to denoise the data effectively, whereas a deep denoiser is able to accurately recover atomic structure of the small platinum nanoparticle and the supporting CeO_2 substrate, which is crucial for the analysis of catalytic dynamics.

B. Supervised denoising

In order to apply the supervised-learning framework described in Section V, we need a training dataset with pairs of noise-free and noisy images. As in many scientific applications, in our case study the only way to generate such a dataset is via computational simulation. This requires prior knowledge of the expected underlying signal structure, a forward model of the imaging system, and a model for the noise.

A simulated dataset relevant to our imaged material of interest is available online [79, 108] (this is the same dataset we use for our experiments on simulated electron-microscopy images). The dataset includes around 100 atomic-scale structural models of platinum nanoparticles with different sizes, shapes and atomic structures on a CeO_2 support with varying

thickness. To simulate these images, the incident electron wave function propagating through the material was generated via the multi-slice method, and then convolved with the point spread function of the electron microscope [57, 7]. These noise-free images are synthetically corrupted by Poisson shot noise to obtain corresponding noisy images. As empirically verified in [79, 108], shot noise dominates other sources of noise such as fixed pattern noise, dark noise, and thermal noise for these data, because the electron detector was operated in electron counting mode at a low electron dose rate per pixel [6].

In the type of phase-contrast TEM imaging relevant to our real data, multiple electron-optical and specimen parameters give rise to complex, nonlinear modulations of the image contrast. These parameters include the objective lens defocus, the thickness and orientation of the specimen, and its crystallographic structure. A reasonable approach to build a training set is to select simulated images with similar contrast to the real data (the type of contrast can be determined by averaging the acquired data). Figure 15 (b) shows the result of training a U-Net using a training set of 5,583 images with similar contrast (more details about the model and training procedure are provided in Appendix F). Unfortunately, the denoised image does not accurately recover the atomic structure of the supporting CeO_2 substrate. As depicted in the zoomed-in

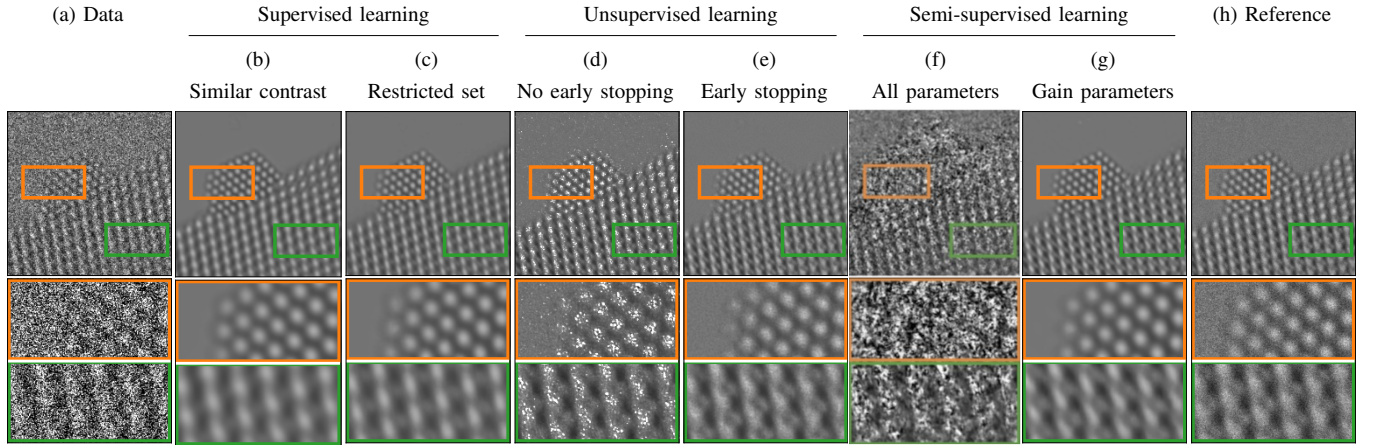


Fig. 15. **Comparison of deep denoisers on real data.** (a) Transmission-electron-microscopy data from the real-world scientific-imaging case study in Section IX, acquired as explained in Appendix F. The data depict a platinum nanoparticle lying on CeO_2 substrate. (b) Denoised estimate obtained via supervised learning using a training set of simulated images with similar contrast to the real data, corrupted with i.i.d. Poisson noise. The result does not recover the atomic structure of the substrate correctly (third row, compared to (h)), the substrate is missing the streaking pattern that descends diagonally from left to right). (c) The substrate structure is recovered correctly (third row, compare to (h)), when the training data are restricted to mimic more closely the parameters of the acquired data. (d) Denoised estimate obtained via unsupervised learning without early stopping. (e) Applying early stopping based on held-out noisy data yields much better results. However, some atoms at the edge of the platinum nanoparticle are not clearly detected (second row, compare to (h)). (f) Finetuning all the parameters in model (b) on the real data results in severe overfitting. (g) Finetuning only the gain parameters produces a high-quality denoised estimate, which recovers the correct atomic structure of both the platinum nanoparticle and the CeO_2 substrate (second and third rows, compare to (h)). (h) Reference image obtained by averaging 40 frames of noisy data.

image on the third row, the substrate is missing the diagonal pattern descending from left to right, which is apparent in a reference image computed by averaging the data (h).

The authors of [108, 79] hypothesized that the failure to accurately recover the substrate structure is due to relative rarity of images with appropriate patterns in the training set. To address this, they selected a smaller subset of 251 images that is painstakingly tailored to the acquisition parameters: structure limited to PtNP2, thickness between 40 Å - 60 Å and, defocus between 5 nm and 10 nm. As shown in Figure 15 (c) the resulting denoised image is able to accurately recover the atomic structure of the substrate. In addition, the denoised intensity values at the edge of the platinum particle, which are key to study its dynamic behavior, matches those of the reference image, as shown in Figure 16. Figure 18 shows examples from the original *similar-contrast* dataset, which have very diverse substrate structures, and from the restricted dataset, where the substrate structure aligns more closely with the observed data. The need to carefully design the training data in order to achieve effective denoising for these data illustrates the importance of training-set design in supervised learning, and the significant domain expertise that it may require in practice.

Figure 10 shows equivalent filters corresponding to several noisy pixels in the noisy data, computed as explained in Section VI. The filters suggest that the network learns to exploit the periodic atomic structure of the CeO_2 substrate and the platinum nanoparticle.

C. Unsupervised denoising

Section IX-B shows that in order to achieve high-quality denoising on our real data of interest via supervised learning, it is necessary to carefully select a restricted simulated training

dataset, which requires extensive technical knowledge of the acquisition parameters. In this section we consider a scenario where such a restricted dataset is not available. The unsupervised learning strategies in Section VII are suitable for such situations, as they do not require a training set with noise-free and noisy signals. Here we apply the blindspot approach described in Section VII-C, utilizing a specialized blindspot U-Net architecture [60], described in more detail in Section F.

Figure 15 (d) shows the result of training the blindspot U-Net on the noisy data. Due to the small number of frames (just 40), the method overfits the noisy data. This can be avoided by applying early stopping based on held-out noisy data, a strategy proposed in [95]. The unsupervised loss (26) is computed on a validation held-out data of 5 images. Training on the remaining 35 images is stopped when the unsupervised loss starts to increase due to overfitting. Figure 15 (e) shows that this greatly improves the denoised estimate. In fact, it correctly recovers the atomic structure of the CeO_2 substrate (third row, compare to (h)). However, it does not clearly detect all the atoms at the edge of the platinum nanoparticle (second row, compare to (h)). This region is particularly challenging to denoise, because its shape is more irregular than the interior regions of the nanoparticle and the substrate, and the limited training data does not allow the unsupervised approach to denoise it sufficiently. This is not surprising because, as explained at the end of Section VII-B and in Figure 12, unsupervised methods tend to require relatively large training sets.

D. Semi-supervised denoising

Section IX-B shows that when we apply supervised learning with the full similar-contrast training set, we correctly identify the atomic structure of the platinum nanoparticle, but not of

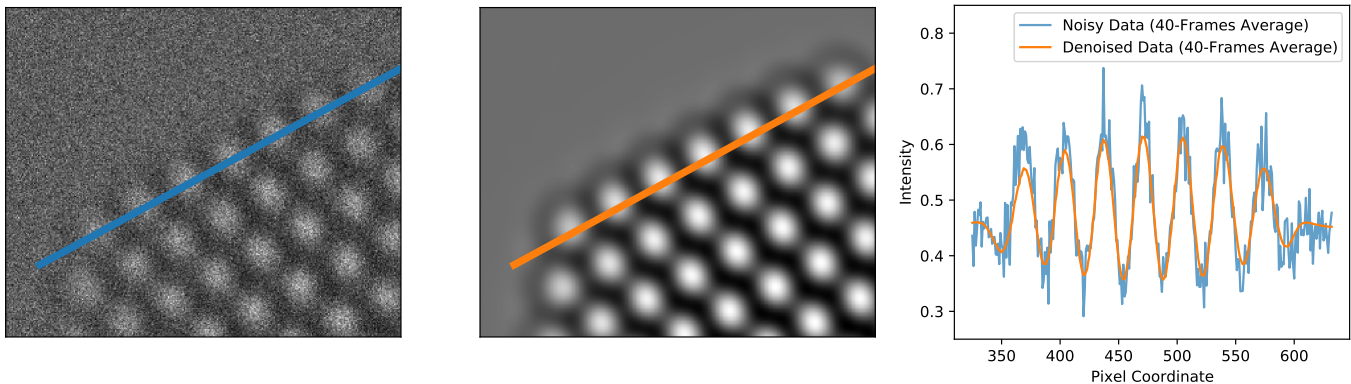


Fig. 16. **Evaluation on real data.** The real dataset in the case study of Section IX consists of 40 frames where the underlying atomic structure is approximately fixed and aligned. Their temporal average therefore provides a reasonable estimate for the underlying intensity profile. The figure shows an evaluation of the supervised denoising model corresponding to Figure 15 (c) from [79]. In the image on the right, we compare the average intensity profile of the atoms situated at the edge of the platinum nanoparticle for the denoised data (middle, orange line) and the temporal average (left, blue line). The profiles are very similar (except for spurious fluctuations in the temporal average), which suggests that the model performs effective denoising on the real data.

the CeO_2 substrate. Conversely, when we apply unsupervised learning, the estimate of CeO_2 substrate is much better, but we fail to detect some atoms at the edge of the platinum nanoparticle. This suggests combining both approaches to achieve the best of both worlds. This can be achieved via a framework known as *semi-supervised learning*.

In this framework, the blindspot U-Net model is first trained in a supervised fashion utilizing the simulated similar-contrast data. Then its parameters are *finetuned* via unsupervised learning on the real data, minimizing the blindspot loss. Figure 15 (f) shows that finetuning all network parameters results in dramatic overfitting. However, finetuning *only the gain parameters in the batch-normalization layers* (see Section IV-B) avoids overfitting, as depicted in Figure 15 (g). This technique is known as GainTuning [78]. The resulting denoised image provides a high-quality reconstruction of the atomic structure of both the platinum nanoparticle and the CeO_2 substrate (compare to (h)).

X. CONCLUSION AND FUTURE DIRECTIONS

In this tutorial we describe how to perform deep denoising using deep convolutional neural networks trained via supervised and unsupervised learning, with an emphasis on considerations that are important for real-world scientific applications. As is apparent in the case study described in Section IX, denoising CNNs achieve impressive results, but their application to real data can be quite challenging. We end by describing several research directions that could contribute to address these challenges and further advance deep denoising methodology.

- In Section IX-B we observe that the performance of supervised denoisers can actually deteriorate when a larger training set is used, because of the presence of additional examples with a different distribution to the data of interest. Developing methodology to systematically design and evaluate simulated training sets in order to optimize performance on real datasets is therefore an important direction for future research.

- The unsupervised learning approaches described in Section VII, which can be applied to individual images (i.e. Neighbor2Neighbor and blindspot methods) critically rely on the assumption that the noise is pixel-wise independent. Developing unsupervised schemes that do not require this assumption would greatly increase the applicability of these methods in practice.
- Vision transformers [45] and diffusion models [47] have achieved impressive results in classification [27], segmentation [56] and image generation [25]. These approaches have begun to be applied to denoising [109, 117, 33, 125, 37, 39, 116, 84], achieving strong results on natural images [65], but their potential remains largely untapped due to the lack of large-scale signal databases in most scientific applications. Such databases could enable the training of general-purpose denoising foundation models, which would require minimum finetuning (as in the semi-supervised framework described in Section IX-D).
- While useful to provide an idea of the strategies learned by deep denoisers, the equivalent-filter interpretation of these models in Section VI is very limited, as it relies on a local linearization of the model. Developing more sophisticated tools to interpret these models would be very useful for practitioners, particularly if they allowed to perform uncertainty quantification.

ACKNOWLEDGEMENTS

We gratefully acknowledge financial support from the National Science Foundation (NSF) through awards OAC-2104105 and OAC-2103936. This work was supported by the NYU IT High Performance Computing resources, services, and staff.

REFERENCES

- [1] Abdelrahman Abdelhamed, Stephen Lin, and Michael S Brown. “A high-quality denoising dataset for smartphone cameras”. In: *Proceedings of the IEEE conference on computer vision and pattern recognition*. 2018, pp. 1692–1700.

- [2] Mehmet Akçakaya et al. “Unsupervised deep learning methods for biological image reconstruction and enhancement: an overview from a signal processing perspective”. In: *IEEE signal processing magazine* 39.2 (2022), pp. 28–44.
- [3] Md Zahangir Alom et al. “The history began from AlexNet: a comprehensive survey on deep learning approaches”. In: *arXiv preprint arXiv:1803.01164* (2018).
- [4] Gilles Aubert and Jean-Francois Aujol. “A variational approach to removing multiplicative noise”. In: *SIAM journal on applied mathematics* 68.4 (2008), pp. 925–946.
- [5] Xiao-Chen Bai, Greg McMullan, and Sjors H.W Scheres. “How cryo-EM is revolutionizing structural biology”. In: *Trends in biochemical sciences* 40.1 (2015), pp. 49–57. DOI: <https://doi.org/10.1016/j.tibs.2014.10.005>.
- [6] Norman Charles Barford. “Experimental measurements: precision, error and truth”. In: (1967).
- [7] Juri Barthel. “Dr. Probe: A software for high-resolution STEM image simulation”. In: *Ultramicroscopy* 193 (2018), pp. 1–11.
- [8] Joshua Batson and Loic Royer. “Noise2self: Blind denoising by self-supervision”. In: *International conference on machine learning*. PMLR. 2019, pp. 524–533.
- [9] Stefan S Biel and Dick Madeley. “Diagnostic virology—the need for electron microscopy: a discussion paper”. In: *Journal of clinical virology* 22.1 (2001), pp. 1–9.
- [10] Yochai Blau and Tomer Michaeli. “The perception-distortion tradeoff”. In: *Proceedings of the IEEE conference on computer vision and pattern recognition*. 2018, pp. 6228–6237.
- [11] Antoni Buades, Bartomeu Coll, and Jean-Michel Morel. “Non-local means denoising”. In: *Image processing on line* 1 (2011), pp. 208–212.
- [12] Harold C Burger, Christian J Schuler, and Stefan Harmeling. “Image denoising: can plain neural networks compete with BM3D?” In: *IEEE conference on computer vision and pattern recognition*. IEEE. 2012, pp. 2392–2399.
- [13] Fernandez-Granda Carlos. *Lectures on mathematical tools for data science*. <https://cds.nyu.edu/wp-content/uploads/2021/05/stationarity-1.pdf>.
- [14] Antonin Chambolle. “An algorithm for total variation minimization and applications”. In: *Journal of mathematical imaging and vision* 20 (2004), pp. 89–97.
- [15] S Grace Chang, Bin Yu, and Martin Vetterli. “Adaptive wavelet thresholding for image denoising and compression”. In: *IEEE Transaction on image processing* 9.9 (2000), pp. 1532–1546.
- [16] Boncellet Charles. “Image noise models”. In: *Handbook of image and video processing* (2005), pp. 397–409.
- [17] Dongdong Chen et al. “Imaging with equivariant deep learning: from unrolled network design to fully unsupervised learning”. In: *IEEE signal processing magazine* 40.1 (2023), pp. 134–147.
- [18] Scott Shaobing Chen, David L Donoho, and Michael A Saunders. “Atomic decomposition by basis pursuit”. In: *SIAM review* 43.1 (2001), pp. 129–159.
- [19] Yunjin Chen and Thomas Pock. “Trainable nonlinear reaction diffusion: A flexible framework for fast and effective image restoration”. In: *IEEE Transactions on pattern analysis and machine intelligence* 39.6 (2016), pp. 1256–1272.
- [20] Peter A Crozier et al. “Visualizing nanoparticle surface dynamics and instabilities enabled by deep denoising”. In: *Science* 387.6737 (2025), pp. 949–954.
- [21] Peter A. Crozier et al. “Dynamic restructuring during processing: approaches to higher temporal resolution”. In: *Microscopy and microanalysis* 25.S2 (2019), pp. 1464–1465.
- [22] Kostadin Dabov et al. “Image denoising by sparse 3-D transform-domain collaborative filtering”. In: *IEEE Transactions on image processing* 16.8 (2007), pp. 2080–2095.
- [23] Kostadin Dabov et al. “Image denoising with block-matching and 3D filtering”. In: *Image processing: algorithms and systems, neural Networks, and machine learning*. Vol. 6064. International Society for Optics and Photonics. 2006, p. 606414.
- [24] Valery Dewil et al. “Self-supervised training for blind multi-frame video denoising”. In: *Proceedings of the IEEE winter conference on applications of computer vision*. 2021, pp. 2724–2734.
- [25] Prafulla Dhariwal and Alexander Nichol. “Diffusion models beat GANs on image synthesis”. In: *Advances in neural information processing systems* 34 (2021), pp. 8780–8794.
- [26] D Donoho and I Johnstone. “Adapting to Unknown Smoothness via Wavelet Shrinkage”. In: *Journal of the American Statistical Association* 90.432 (Dec. 1995).
- [27] Alexey Dosovitskiy et al. “An image is worth 16x16 words: transformers for image recognition at scale”. In: *arXiv preprint arXiv:2010.11929* (2020).
- [28] Thibaud Ehret et al. “Model-Blind Video Denoising via Frame-To-Frame Training”. In: *Proceedings of the IEEE conference on computer vision and pattern recognition*. 2019, pp. 11361–11370.
- [29] Michael Elad and Michal Aharon. “Image denoising via sparse and redundant representations over learned dictionaries”. In: *IEEE Transaction on image processing* 15.12 (2006), pp. 3736–3745.
- [30] Michael Elad, Bahjat Kowar, and Gregory Vaksman. “Image Denoising: The Deep Learning Revolution and Beyond—A Survey Paper—”. In: *arXiv preprint arXiv:2301.03362* (2023).
- [31] Michael Elad, Peyman Milanfar, and Ron Rubinstein. “Analysis versus synthesis in signal priors”. In: *Inverse problems* 23.3 (2007), p. 947.
- [32] Peter Ercius et al. “The 4D Camera – a 87 kHz Frame-rate Detector for Counted 4D-STEM Experiments”. In:

- Microscopy and microanalysis* (2020), pp. 1–3. DOI: 10.1017/S1431927620019753.
- [33] Chi-Mao Fan, Tsung-Jung Liu, and Kuan-Hsien Liu. “SUNet: swin transformer UNet for image denoising”. In: *IEEE international symposium on circuits and systems*. IEEE. 2022, pp. 2333–2337.
 - [34] Linwei Fan et al. “Brief review of image denoising techniques”. In: *Visual computing for industry, biomedicine, and art 2.1* (2019), pp. 1–12.
 - [35] A.R. Faruqi and G. McMullan. “Direct imaging detectors for electron microscopy”. In: *Nuclear instruments and methods in physics research section A: accelerators, spectrometers, detectors and associated Equipment 878* (2018). Radiation Imaging Techniques and Applications, pp. 180–190. DOI: <https://doi.org/10.1016/j.nima.2017.07.037>.
 - [36] Alessandro Foi et al. “Practical Poissonian-Gaussian noise modeling and fitting for single-image raw-data”. In: *IEEE Transactions on image processing* 17.10 (2008), pp. 1737–1754.
 - [37] Qi Gao and Hongming Shan. “CoCoDiff: a contextual conditional diffusion model for low-dose CT image denoising”. In: *Developments in X-Ray tomography XIV*. Vol. 12242. SPIE. 2022.
 - [38] Xavier Glorot, Antoine Bordes, and Yoshua Bengio. “Deep sparse rectifier neural networks”. In: *Proceedings of the international conference on artificial intelligence and statistics*. 2011, pp. 315–323.
 - [39] Kuang Gong et al. “PET image denoising based on denoising diffusion probabilistic model”. In: *European journal of nuclear medicine and molecular imaging* (2023), pp. 1–11.
 - [40] Ian Goodfellow, Yoshua Bengio, and Aaron Courville. *Deep learning*. MIT press, 2016.
 - [41] Bhawna Goyal et al. “Image denoising review: From classical to state-of-the-art approaches”. In: *Information fusion* 55 (2020), pp. 220–244.
 - [42] Shuhang Gu and Radu Timofte. “A brief review of image denoising algorithms and beyond”. In: *Inpainting and denoising challenges* (2019), pp. 1–21.
 - [43] Shuhang Gu et al. “Weighted nuclear norm minimization with application to image denoising”. In: *Proceedings of the IEEE conference on computer vision and pattern recognition*. 2014, pp. 2862–2869.
 - [44] Bichuan Guo et al. “Learning Model-Blind Temporal Denoisers without Ground Truths”. In: *arXiv preprint arXiv:2007.03241* (2020).
 - [45] Kai Han et al. “A survey on vision transformer”. In: *IEEE Transactions on pattern analysis and machine intelligence* 45.1 (2022), pp. 87–110.
 - [46] Y Hel-Or and D Shaked. “A discriminative approach for wavelet denoising”. In: *IEEE Transaction on image processing* (2008).
 - [47] Jonathan Ho, Ajay Jain, and Pieter Abbeel. “Denoising diffusion probabilistic models”. In: *Advances in neural information processing systems* 33 (2020), pp. 6840–6851.
 - [48] Timothy O Hodson, Thomas M Over, and Sydney S Foks. “Mean squared error, deconstructed”. In: *Journal of advances in modeling earth systems* 13.12 (2021), e2021MS002681.
 - [49] Tao Huang et al. “Neighbor2neighbor: Self-supervised denoising from single noisy images”. In: *Proceedings of the IEEE conference on computer vision and pattern recognition*. 2021, pp. 14781–14790.
 - [50] Ademola E Ilesanmi and Taiwo O Ilesanmi. “Methods for image denoising using convolutional neural network: a review”. In: *Complex & intelligent systems* 7.5 (2021), pp. 2179–2198.
 - [51] Sergey Ioffe and Christian Szegedy. “Batch normalization: Accelerating deep network training by reducing internal covariate shift”. In: *International conference on machine learning*. pmlr. 2015, pp. 448–456.
 - [52] Xixi Jia et al. “FOCNet: a fractional optimal control network for image denoising”. In: *Proceedings of the IEEE conference on computer vision and pattern recognition*. 2019, pp. 6054–6063.
 - [53] Ulugbek S Kamilov et al. “Plug-and-play methods for integrating physical and learned models in computational imaging: theory, algorithms, and applications”. In: *IEEE signal processing magazine* 40.1 (2023), pp. 85–97.
 - [54] Wesley Khademi et al. “Self-Supervised Poisson-Gaussian Denoising”. In: *Proceedings of the IEEE winter conference on applications of computer vision*. 2021, pp. 2131–2139.
 - [55] Diederik P Kingma and Jimmy Ba. “Adam: A method for stochastic optimization”. In: *arXiv preprint arXiv:1412.6980* (2014).
 - [56] Alexander Kirillov et al. “Segment anything”. In: *Proceedings of the IEEE international conference on computer vision*. 2023, pp. 4015–4026.
 - [57] Earl J Kirkland et al. “Image simulation in transmission electron microscopy”. In: *Cornell University, Ithaca* (2006).
 - [58] Alexander Krull, Tim-Oliver Buchholz, and Florian Jug. “Noise2Void - Learning Denoising From Single Noisy Images”. In: *Proceedings of the IEEE conference on computer vision and pattern recognition*. 2019, pp. 2124–2132.
 - [59] Alexander Krull, Tomas Vicar, and Florian Jug. “Probabilistic Noise2Void: unsupervised content-aware denoising”. In: *arXiv preprint arXiv:1906.00651* (2019).
 - [60] Samuli Laine et al. “High-Quality Self-Supervised Deep Image Denoising”. In: *Advances in neural information processing systems*. 2019, pp. 6970–6980.
 - [61] Ethan L. Lawrence et al. “Approaches to exploring spatio-temporal surface dynamics in nanoparticles with in situ transmission electron microscopy”. In: *Microscopy and microanalysis* 26.1 (2020), pp. 86–94. DOI: 10.1017/S1431927619015228.
 - [62] Yann LeCun, Yoshua Bengio, et al. “Convolutional networks for images, speech, and time series”. In: *The handbook of brain theory and neural networks* 3361.10 (1995), p. 1995.

- [63] Yann LeCun, Yoshua Bengio, and Geoffrey Hinton. “Deep learning”. In: *nature* 521.7553 (2015), pp. 436–444.
- [64] Jaakko Lehtinen et al. “Noise2Noise: Learning Image Restoration without Clean Data”. In: *International conference on machine learning*. 2018, pp. 2965–2974.
- [65] Jingyun Liang et al. “SwinIR: Image restoration using Swin transformer”. In: *Proceedings of the IEEE international conference on computer vision*. 2021, pp. 1833–1844.
- [66] Jeff W Lichtman and José-Angel Conchello. “Fluorescence microscopy”. In: *Nature methods* 2.12 (2005), pp. 910–919.
- [67] Ding Liu et al. “Non-local recurrent network for image restoration”. In: *Advances in neural information processing systems* 31 (2018).
- [68] Florian Luisier, Thierry Blu, and Michael Unser. “A new SURE approach to image denoising: Interscale orthonormal wavelet thresholding”. In: *IEEE Transactions on image processing* 16.3 (2007), pp. 593–606.
- [69] Florian Luisier, Thierry Blu, and Michael Unser. “Image denoising in mixed Poisson–Gaussian noise”. In: *IEEE Transactions on image processing* 20.3 (2010), pp. 696–708.
- [70] Julien Mairal, Francis Bach, Jean Ponce, et al. “Sparse modeling for image and vision processing”. In: *Foundations and trends® in computer graphics and vision* 8.2-3 (2014), pp. 85–283.
- [71] Lukasz Malinski, Krystian Radlak, and Bogdan Smolka. “Is large improvement in efficiency of impulsive noise removal in color images still possible?” In: *Plos one* 16.6 (2021), e0253117.
- [72] Stéphane Mallat. *A wavelet tour of signal processing*. Elsevier, 1999.
- [73] Adria Marcos-Morales et al. “Evaluating Unsupervised Denoising Requires Unsupervised Metrics”. In: *International conference on machine learning* (2023).
- [74] D. Martin et al. “A database of human segmented natural images and its application to evaluating segmentation algorithms and measuring ecological statistics”. In: *Proceedings of the IEEE international conference on computer vision*. Vol. 2. July 2001, pp. 416–423.
- [75] Ian S McLean. *Electronic imaging in astronomy: detectors and instrumentation*. Springer Science & Business Media, 2008.
- [76] Christopher A Metzler et al. “Unsupervised Learning with Stein’s Unbiased Risk Estimator”. In: *arXiv preprint arXiv:1805.10531* (2018).
- [77] Peyman Milanfar. “A tour of modern image filtering: New insights and methods, both practical and theoretical”. In: *IEEE signal processing magazine* 30.1 (2012), pp. 106–128.
- [78] Sreyas Mohan et al. “Adaptive denoising via gaintuning”. In: *Advances in neural information processing systems* 34 (2021), pp. 23727–23740.
- [79] Sreyas Mohan et al. “Deep denoising for scientific discovery: a case study in electron microscopy”. In: *IEEE Transactions on computational imaging* (2022).
- [80] Sreyas Mohan et al. “Robust and interpretable blind image denoising via bias-free convolutional neural networks”. In: *International conference on learning representations*. 2020.
- [81] Grégoire Montavon et al. “Explaining nonlinear classification decisions with deep taylor decomposition”. In: *Pattern recognition* 65 (2017), pp. 211–222.
- [82] Tiziano Montini et al. “Fundamentals and catalytic applications of CeO₂-based materials”. In: *Chemical reviews* 116.10 (2016), pp. 5987–6041.
- [83] Yao Nie, Li Li, and Zidong Wei. “Recent advancements in Pt and Pt-free catalysts for oxygen reduction reaction”. In: *Chemical society reviews* 44.8 (2015), pp. 2168–2201.
- [84] Naama Pearl et al. “SVNR: spatially-variant noise removal with denoising diffusion”. In: *arXiv preprint arXiv:2306.16052* (2023).
- [85] Luke Pfister and Yoram Bresler. “Learning filter bank sparsifying transforms”. In: *IEEE Transactions on signal processing* 67.2 (2018), pp. 504–519.
- [86] Tobias Plotz and Stefan Roth. “Benchmarking denoising algorithms with real photographs”. In: *Proceedings of the IEEE conference on computer vision and pattern recognition*. 2017, pp. 1586–1595.
- [87] Mangal Prakash et al. “Fully unsupervised probabilistic Noise2Void”. In: *arXiv preprint arXiv:1911.12291* (2019).
- [88] M Raphan and E P Simoncelli. “Optimal denoising in redundant representations”. In: *IEEE Transaction on image processing* 17.8 (Aug. 2008), pp. 1342–1352. DOI: 10.1109/TIP.2008.925392.
- [89] Saiprasad Ravishankar and Yoram Bresler. “Learning sparsifying transforms”. In: *IEEE Transactions on signal processing* 61.5 (2012), pp. 1072–1086.
- [90] Saiprasad Ravishankar and Yoram Bresler. “Sparsifying transform learning with efficient optimal updates and convergence guarantees”. In: *IEEE Transactions on signal processing* 63.9 (2015), pp. 2389–2404.
- [91] Robin Rombach et al. “High-resolution image synthesis with latent diffusion models”. In: *Proceedings of the IEEE conference on computer vision and pattern recognition*. 2022, pp. 10684–10695.
- [92] Olaf Ronneberger, Philipp Fischer, and Thomas Brox. “U-Net: convolutional networks for biomedical image segmentation”. In: *International conference on medical image computing and computer-assisted intervention*. Springer. 2015, pp. 234–241.
- [93] Leonid I Rudin and Stanley Osher. “Total variation based image restoration with free local constraints”. In: *Proceedings of international conference on image processing*. Vol. 1. IEEE. 1994, pp. 31–35.
- [94] José M Sancho et al. “Analytical and numerical studies of multiplicative noise”. In: *Physical review A* 26.3 (1982), p. 1589.
- [95] Dev Yashpal Sheth et al. “Unsupervised deep video denoising”. In: *Proceedings of the IEEE international conference on computer vision*. 2021, pp. 1759–1768.

- [96] E P Simoncelli and E H Adelson. “Noise removal via Bayesian wavelet coring”. In: *Proceeding of the IEEE International conference on image processing*. Vol. I. Lausanne, Sept. 1996, pp. 379–382.
- [97] Eero P Simoncelli and William T Freeman. “The steerable pyramid: A flexible architecture for multi-scale derivative computation”. In: *Proceedings., International Conference on Image Processing*. Vol. 3. IEEE. 1995, pp. 444–447.
- [98] Karen Simonyan, Andrea Vedaldi, and Andrew Zisserman. “Deep inside convolutional networks: Visualising image classification models and saliency maps”. In: *arXiv preprint arXiv:1312.6034* (2013).
- [99] David Smith. “Chapter 1: Characterization of nanomaterials using transmission electron microscopy”. English (US). In: *Hierarchical Nanostructures for Energy Devices*. Ed. by Angus I. Kirkland and Sarah J. Haigh. 37th ed. RSC Nanoscience and Nanotechnology 37. Royal Society of Chemistry, Jan. 2015, pp. 1–29. DOI: 10.1039/9781782621867-00001.
- [100] Shakarim Soltanayev and Se Young Chun. “Training deep learning based denoisers without ground truth data”. In: *Advances in neural information processing systems*. Vol. 31. 2018.
- [101] Jean-Luc Starck, Emmanuel J Candès, and David L Donoho. “The curvelet transform for image denoising”. In: *IEEE Transactions on image processing* 11.6 (2002), pp. 670–684.
- [102] Hiroyuki Takeda, Sina Farsiu, and Peyman Milanfar. “Kernel regression for image processing and reconstruction”. In: *IEEE Transactions on image processing* 16.2 (2007), pp. 349–366.
- [103] Franklin Tao and Peter Crozier. “Atomic-scale observations of catalyst structures under reaction conditions and during catalysis”. In: *Chemical reviews* 116.6 (Mar. 2016), pp. 3487–3539. DOI: 10.1021/cr5002657.
- [104] Rini Smita Thakur, Ram Narayan Yadav, and Lalita Gupta. “State-of-art analysis of image denoising methods using convolutional neural networks”. In: *IET image processing* 13.13 (2019), pp. 2367–2380.
- [105] Chunwei Tian et al. “Deep learning on image denoising: An overview”. In: *Neural networks* 131 (2020), pp. 251–275.
- [106] Carlo Tomasi and Roberto Manduchi. “Bilateral filtering for gray and color images.” In: *Proceedings of the IEEE international conference on computer vision*. Vol. 98. 1. 1998.
- [107] Dmitry Ulyanov, Andrea Vedaldi, and Victor Lempitsky. “Deep image prior”. In: *Proceedings of the IEEE conference on computer vision and pattern recognition*. 2018, pp. 9446–9454.
- [108] Joshua L Vincent et al. “Developing and evaluating deep neural network-based denoising for nanoparticle tem images with ultra-low signal-to-noise”. In: *Microscopy and microanalysis* 27.6 (2021), pp. 1431–1447.
- [109] Dayang Wang, Zhan Wu, and Hengyong Yu. “Tedenet: Convolution-free T2T vision transformer-based encoder-decoder dilation network for low-dose CT denoising”. In: *Machine learning in medical imaging*. Springer. 2021, pp. 416–425.
- [110] Zhong Lin Wang. “New developments in transmission electron microscopy for nanotechnology”. In: *Advanced materials* 15.18 (2003), pp. 1497–1514.
- [111] Zhou Wang et al. “Image quality assessment: from error visibility to structural similarity”. In: *IEEE Transactions on image processing* 13.4 (2004), pp. 600–612.
- [112] Bihan Wen, Saiprasad Ravishankar, and Yoram Bresler. “FRIST—Flipping and rotation invariant sparsifying transform learning and applications”. In: *Inverse problems* 33.7 (2017), p. 074007.
- [113] Bihan Wen, Saiprasad Ravishankar, and Yoram Bresler. “Structured overcomplete sparsifying transform learning with convergence guarantees and applications”. In: *International journal of computer vision* 114.2 (2015), pp. 137–167.
- [114] Norbert Wiener. *Extrapolation, interpolation, and smoothing of stationary time series: with engineering applications*. The MIT press, 1949.
- [115] Jun Xu et al. “Real-world noisy image denoising: A new benchmark”. In: *arXiv preprint arXiv:1804.02603* (2018).
- [116] Cheng Yang, Lijing Liang, and Zhixun Su. “Real-world denoising via diffusion model”. In: *arXiv preprint arXiv:2305.04457* (2023).
- [117] Chao Yao et al. “Dense residual Transformer for image denoising”. In: *Electronics* 11.3 (2022), p. 418.
- [118] Songhyun Yu et al. “Joint Learning of Blind Video Denoising and Optical Flow Estimation”. In: *Proceedings of the IEEE conference on computer vision and pattern recognition Workshops*. June 2020.
- [119] Weiting Yu, Marc D Porosoff, and Jingguang G Chen. “Review of Pt-based bimetallic catalysis: from model surfaces to supported catalysts”. In: *Chemical reviews* 112.11 (2012), pp. 5780–5817.
- [120] Kai Zhang et al. “Beyond a Gaussian denoiser: Residual learning of deep CNN for image denoising”. In: *IEEE Transactions on image processing* 26.7 (2017), pp. 3142–3155.
- [121] Richard Zhang et al. “The unreasonable effectiveness of deep features as a perceptual metric”. In: *Proceedings of the IEEE conference on computer vision and pattern recognition*. 2018, pp. 586–595.
- [122] Xiaoshuai Zhang et al. “Dynamically unfolding recurrent restorer: A moving endpoint control method for image restoration”. In: *arXiv preprint arXiv:1805.07709* (2018).
- [123] Yide Zhang et al. “A Poisson-Gaussian denoising dataset with real fluorescence microscopy images”. In: *Proceedings of the IEEE conference on computer vision and pattern recognition*. 2019, pp. 11710–11718.
- [124] Hang Zhao et al. “Loss functions for image restoration with neural networks”. In: *IEEE Transactions on computational imaging* 3.1 (2016), pp. 47–57.

- [125] Mo Zhao et al. “Hybrid transformer-CNN for real image denoising”. In: *IEEE signal processing letters* 29 (2022), pp. 1252–1256.
- [126] Magaiya Zhussip, Shakarim Soltanayev, and Se Young Chun. “Extending Stein’s unbiased risk estimator to train deep denoisers with correlated pairs of noisy images”. In: *Advances in neural information processing systems* 32 (2019).
- [127] Jian-Min Zuo and J.C.H. Spence. *Advanced transmission electron microscopy, imaging and diffraction in nanoscience*. Jan. 2017. ISBN: ISBN 978-1-4939-6607-3.
- [128] Louise Zylberberg and Michel Laurin. “Analysis of fossil bone organic matrix by transmission electron microscopy”. In: *Comptes rendus palevol* 10.5-6 (2011), pp. 357–366.

Sreyas Mohan (sreyasmohan@meta.com) is a Senior Research Scientist with Meta Platforms Inc. His research focuses on deep learning, image processing and computer vision. He received his Ph.D. from New York University in 2022, and B.Tech from Indian Institute of Technology Madras in 2017.

Kangning Liu (kl3141@nyu.edu) is a Research Scientist with Adobe Inc. He received his Ph.D. from New York University in 2024. He received a B.E. in Electrical Engineering at Tsinghua University in 2017, and an M.Sc. in Data Science at ETH Zurich in 2019. In 2022 and 2023, he served as a research intern at Google and Adobe, respectively. His research focuses on deep learning and computer vision.

Peter A. Crozier (crozier@asu.edu) is a Professor of Materials in the School for Engineering of Matter, Transport and Energy at Arizona State University. He has extensive experience in developing and applying advanced transmission electron microscopy techniques to problems related to energy and the environment with special emphasis on electroceramics and catalytic materials. He is currently interested in exploring the application of newly emerging data science tools to understand atomic level structural dynamics in materials.

Carlos Fernandez-Granda (cfgranda@cims.nyu.edu) is an Associate Professor at the Courant Institute of Mathematical Sciences and the Center for Data Science in New York University. His research focuses on the design and analysis of data-science methodology, with a particular focus on machine learning, and its application to medicine, climate science and scientific imaging.

APPENDIX

A. Example dataset of 1D piecewise-constant signals

1) *Dataset*: The signals are simulated by generating *jumps* at random with probability 0.3, and constraining the separation between the jumps to be at least 5. The signal amplitude between jumps is chosen uniformly at random between 0 and 1. The total length of each signal is 100. In order to simulate the noisy data, the noise-free signals were corrupted with additive i.i.d. Gaussian noise with standard deviation equal to 0.1.

We created a training set with 500 signals and a test set with 100 signals. In some experiments we utilize smaller training sets, as indicated in the corresponding caption.

2) *Model*: For all experiments we used a convolutional neural network with 10 convolutional layers (except when indicated otherwise) interleaved with ReLU layers. Each of these layers consists of 64 1D filters each with a kernel size of 3 (i.e., each hidden layer has 64 channels). Unless indicated otherwise, a skip connection connects the model input to the output layer.

B. Example dataset of natural images

1) *Dataset*: We use 180×180 natural images from the Berkeley Segmentation Dataset [74]. The training and test sets comprise 400 and 68 images respectively. The training set is augmented via downsampling, random flips, and random rotations of patches in these images [120, 80]. We train the DnCNN on patches of size 50×50 , which yields a total of 541,600 noise-free training patches, and the U-Net architecture we use patches of size 128×128 for a total of 22,400 training patches. To generate the noisy data we corrupt the images with i.i.d. additive Gaussian noise with a standard deviation sampled uniformly between 0 and 100 separately for each image (with pixels values in 0 to 255 range).

2) *Models*: We employed the following models, depicted in Figure 17, in our experiments:

- **DnCNN**. DnCNN [120] consists of 20 convolutional layers, each consisting of 3×3 filters and 64 channels, batch normalization [51], and a ReLU nonlinearity. It has a skip connection from the initial layer to the final layer, which has no nonlinear units. We use the pre-trained weights released by the authors. For the bias-free version of this architecture, **BFCNN**, where additive bias from every layer is removed, we use the pre-trained weight released by authors of [80].
- **U-Net**. The U-Net model [92] has the following structure:
 - 1) *conv1* - Takes in input image and maps to 32 channels with 5×5 convolutional kernels.
 - 2) *conv2* - Input: 32 channels. Output: 32 channels. 3×3 convolutional kernels.
 - 3) *conv3* - Input: 32 channels. Output: 64 channels. 3×3 convolutional kernels with stride 2.
 - 4) *conv4* - Input: 64 channels. Output: 64 channels. 3×3 convolutional kernels.
 - 5) *conv5* - Input: 64 channels. Output: 64 channels. 3×3 convolutional kernels with dilation factor of 2.

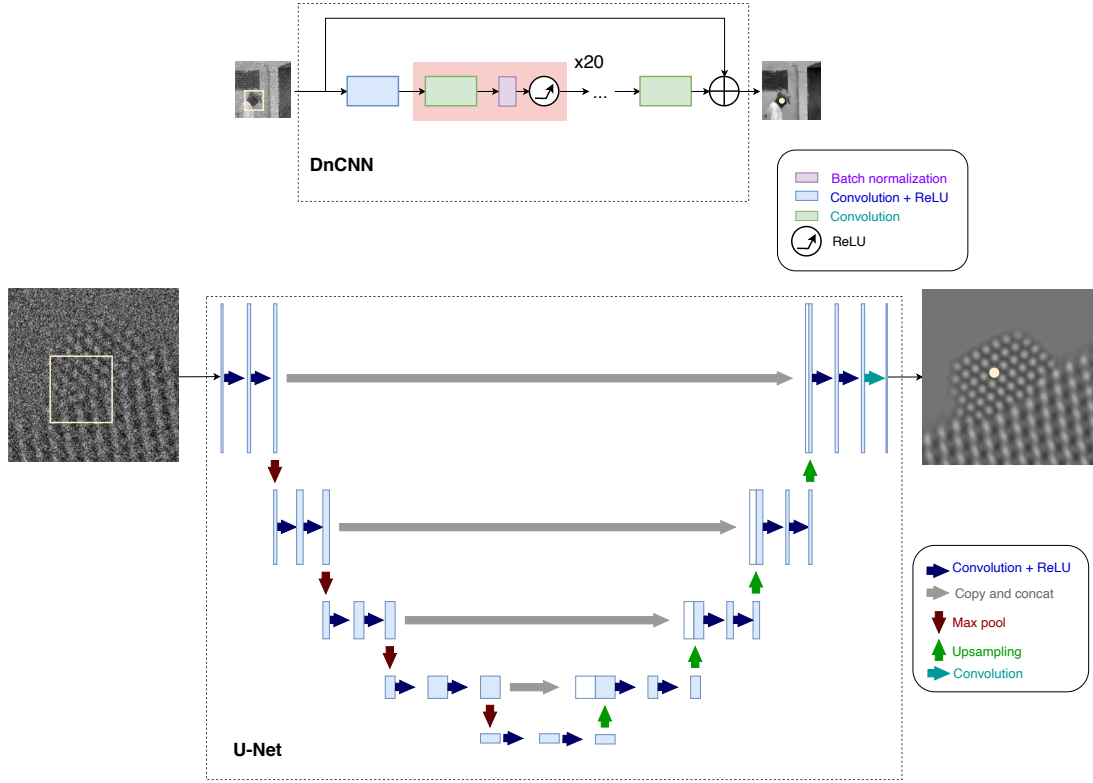


Fig. 17. **DnCNN and U-Net architectures.** The diagram depicts the different components of the DnCNN and U-Net neural networks.

- 6) *conv6*- Input: 64 channels. Output: 64 channels. 3×3 convolutional kernels with dilation factor of 4.
- 7) *conv7*- Transposed convolution layer. Input: 64 channels. Output: 64 channels. 4×4 filters with stride 2.
- 8) *conv8*- Input: 96 channels. Output: 64 channels. 3×3 convolutional kernels. The input to this layer is the concatenation of the outputs of layer *conv7* and *conv2*.
- 9) *conv9*- Input: 32 channels. Output: 1 channels. 5×5 convolutional kernels.

We use pretrained weights released by the authors of [80].

C. Example dataset of simulated electron-microscopy images

1) *Dataset*: To create this dataset we selected a subset of the simulated transmission-electron-microscopy dataset of platinum nanoparticles released in [79]. The training and test sets consist of 353 and 19 images respectively.

2) *Model*: The model for electron-microscopy images (unless indicated otherwise) is the U-Net architecture proposed in [79], with $n = 6$ scales and 128 base channels. The network consists of n down-blocks and n up-blocks. A down-block consists of a max-pooling layer, which reduces the spatial-dimension by half, followed by a *conv-block*. Similarly, an up-block consists of bilinear upsampling, which enlarges the size of the feature-map by a factor of two, followed by *conv-block*. A *conv-block* consists of conv-BN-ReLU-conv-BN-ReLU, where conv represents a convolutional layer and BN stands for batch normalization [51].

D. Supervised learning on example datasets

We trained all models over 100 epochs using the Adam optimizer [55] to minimize the mean squared error between the noise-free and noisy images, with an initial learning rate of 0.001, adjusted by a learning rate scheduler. We utilized continuous noise sampling, generating fresh noisy data corresponding to each noise-free signal in the training set at each epoch.

E. Unsupervised learning on example datasets

We performed unsupervised learning following the Noise2Noise framework [64], minimizing the unsupervised cost function in (22), where the target is generated by adding independent noise to the same noise-free signal. The training parameters are otherwise the same as in the supervised models.

F. Real electron microscopy data

1) *Dataset*: The real dataset contains 40 TEM images of platinum nanoparticles on a CeO_2 substrate, synthesized through standard hydrothermal and metal deposition methods. The material was prepared by dispersing platinum/ CeO_2 powder onto a windowed micro electro-mechanical system (MEMS)-based Si_3N_4 chip. After loading the sample into the ETEM, nitrogen gas was leaked into the cell until an ambient pressure of 5×10^{-3} Torr N_2 was achieved; the temperature was maintained at 20°C . The data were acquired on an aberration-corrected FEI Titan ETEM operated at 300 kV. Time-resolved image series were captured using a Gatan K2

IS direct electron detector at a speed of 40 frames per second, yielding a time resolution of 25 milliseconds per frame. An incident electron beam dose rate of $5,000 \text{ e}^-/\text{\AA}^2/\text{s}$ was used; for the pixel size employed during the experiment (i.e., $0.061 \text{ \AA}/\text{pixel}$), these conditions resulted in an average dose of $0.45 \text{ e}^-/\text{pixel}/\text{frame}$. The electron beam was blanked when images were not being acquired. We refer the reader to [79, 108] for more details.

2) *Models*: We used U-Nets [92] with large fields of view for our experiments with real TEM images. For supervised learning, we used the U-Net architecture described in Section C2. For unsupervised learning, we used a similar U-Net architecture, but with an architecturally imposed blind-spot as introduced in Ref. [95]. For the supervised, unsupervised and semi-supervised models, we use the trained weights released by [79, 95, 78] respectively.

G. Derivation of the Wiener Filter in the frequency domain

Following the notation in Section III, we represent the noise-free signal by the random vector $\tilde{x}$, the noisy data by the random vector $\tilde{y}$ and the convolutional filter by w . These signals can be multidimensional. By the convolution theorem, the discrete Fourier transform (DFT) $\mathcal{F}$ of the convolution between two signals is proportional to the elementwise product $\odot$ of their DFTs, so

$$\|\tilde{x} - w * \tilde{y}\|^2 \propto \|\mathcal{F}\tilde{x} - \mathcal{F}(w * \tilde{y})\|^2 \quad (27)$$

$$= \|\mathcal{F}\tilde{x} - \mathcal{F}w \odot \mathcal{F}\tilde{y}\|^2 \quad (28)$$

$$= \sum_{k=1}^K (\mathcal{F}\tilde{x}[k] - \mathcal{F}w[k] \mathcal{F}\tilde{y}[k])^2, \quad (29)$$

where K is the number of DFT coefficients in each signal. Consequently, minimizing the MSE can be achieved by separately minimizing each frequency component, since

$$\arg \min_w \mathbb{E} [\|\tilde{x} - w * \tilde{y}\|^2] \quad (30)$$

$$= \arg \min_w \sum_{k=1}^K (\mathcal{F}\tilde{x}[k] - \mathcal{F}w[k] \mathcal{F}\tilde{y}[k])^2, \quad (31)$$

which can be minimized by selecting $\mathcal{F}w[k]$ to equal

$$\arg \min_f \mathbb{E} [(\mathcal{F}\tilde{x}[k] - f \cdot \mathcal{F}\tilde{y}[k])^2]. \quad (32)$$

In this derivation we assume that the signal is periodic, or equivalently that it *wraps* around. In practice this is often not the case, so care must be taken to deal with border effects (e.g. via zero padding or mirroring).

H. Analysis of unsupervised learning loss

By linearity of expectation

$$\mathbb{E} \left[\left(\tilde{t}_j - \mathcal{D}(\tilde{y})_j \right)^2 \right] = \mathbb{E} \left[\left(\tilde{t}_j - x_j + x_j - \mathcal{D}(\tilde{y})_j \right)^2 \right] \quad (33)$$

$$= \mathbb{E} \left[\left(\tilde{t}_j - x_j \right)^2 \right] + \mathbb{E} \left[\left(x_j - \mathcal{D}(\tilde{y})_j \right)^2 \right] \quad (34)$$

$$+ 2\mathbb{E} \left[\left(\tilde{t}_j - x_j \right) \left(x_j - \mathcal{D}(\tilde{y})_j \right) \right] \quad (35)$$

$$= \mathbb{E} \left[\left(\tilde{t}_j - x_j \right)^2 \right] + \mathbb{E} \left[\left(x_j - \mathcal{D}(\tilde{y})_j \right)^2 \right]. \quad (36)$$

The last equality relies on the assumptions that $\mathbb{E}[\tilde{t}_j] = x_j$ and that $\tilde{t}_j$ and $\tilde{y}_j$ are independent, which implies that $\tilde{t}_j$ and $\mathcal{D}(\tilde{y})_j$ are also independent, so that

$$\mathbb{E} \left[\left(\tilde{t}_j - x_j \right) \left(x_j - \mathcal{D}(\tilde{y})_j \right) \right] = (\mathbb{E}[\tilde{t}_j] - x_j) \left(x_j - \mathbb{E}[\mathcal{D}(\tilde{y})_j] \right) = 0. \quad (37)$$

I. Likelihood map

The likelihood map quantifies the consistency of two alternative hypotheses with the observed data. The hypotheses concern the underlying signal of interest, and are established based on the denoised estimate. In Figure 13 the hypotheses are whether the detected atoms are (1) truly there, or (2) spurious artifacts generated by the denoiser due to the noise in the data.

In order to compute the likelihood of the data under a certain hypothesis we follow the following steps:

- 1) A region of interest $\mathcal{R}$ in the denoised signal is identified. For the example in Figure 13, this corresponds to an atom, located by some detection algorithm.
- 2) A low-dimensional model is fit to the denoised image within the region of interest $\mathcal{R}$. The low-dimensional model provides an estimate of the signal value $\hat{x}_i$ at each entry $i \in \mathcal{R}$. For the two hypotheses relevant to Figure 13, which posit the presence or absence of an atom, the intensity of the atom and the vacuum are assumed to be constant. For hypothesis 1 (presence of an atom), the estimate is obtained by averaging over all denoised pixels in $\mathcal{R}$. For hypothesis 2 (absence of an atom), the estimate is obtained by averaging over all denoised pixels in the vacuum region, where no atoms are detected.
- 3) The likelihood of the noisy data within $\mathcal{R}$ is computed, assuming that each hypothesis holds. This requires characterizing the distribution of the noisy data with respect to the underlying signal. In Figure 13, the noise is well approximated as independent and individually distributed Poisson (see Section 4.3 in [79]), so the likelihood is

$$\mathcal{L}(\mathcal{R}) := \prod_{i \in \mathcal{R}} p_{\hat{x}_i}(y_i), \quad (38)$$

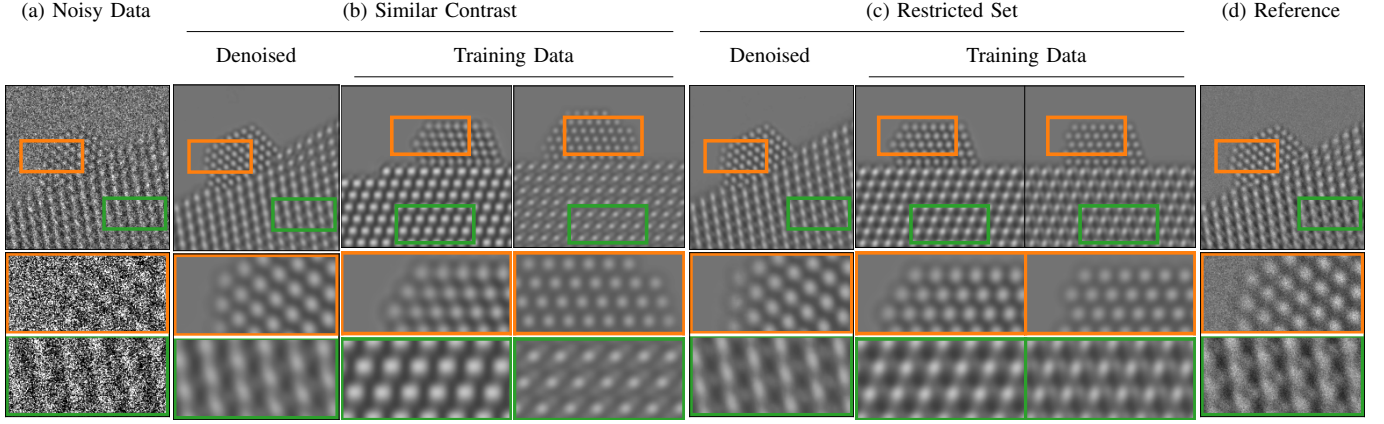


Fig. 18. **Influence of training data in supervised deep denoising.** (a) Transmission-electron-microscopy data from the real-world scientific-imaging case study in Section IX, acquired as explained in Appendix F. The data depict a platinum nanoparticle lying on CeO_2 substrate. (b) Denoising results obtained via supervised learning on a training set simulated to have similar contrast as the observed data, using a relatively wide range of imaging parameters. Two training examples are also shown, which illustrate the resulting diversity of patterns in the CeO_2 substrate (third row). Denoising does not recover the structure of the substrate correctly (third row, compared to (d)), the substrate is missing the streaking pattern that descends diagonally from left to right). (c) Denoising results obtained via supervised learning on a subset of the similar-contrast training set restricted to approximate the parameters of the acquired data more closely. Two training examples are also shown, which illustrate that the patterns in the CeO_2 substrate are less diverse and more aligned with the observed data (third row, compare to (b) and (d)). Denoising does recover the substrate structure (third row, compare to (d)). (d) Reference image obtained by averaging 40 frames of noisy data.

where y_i denotes the noisy value in the i th pixel, and $p_{\hat{x}_i}$ is a Poisson probability mass function with rate parameter $\hat{x}_i$. In the low-dimensional model that assumes constant intensities, $\hat{x}_i$ is constant for all pixels in $\mathcal{R}$.

The likelihood map is equal to the difference between the logarithm of the likelihoods corresponding to the two competing hypotheses for each region of interest. This quantifies the relative consistency of the two hypotheses with the measured data.

J. Influence of training data in supervised denoising

Figure 18 illustrates how the design of the training data in supervised deep denoising can affect the denoising results. The figure shows the transmission-electron-microscopy data from the real-world scientific-imaging case study in Section IX (a), which depict a platinum nanoparticle lying on CeO_2 substrate. As explained in Section IX-B, a supervised deep denoiser trained on a set of images simulated to have similar contrast as the observed data does not recover the atomic structure of the substrate correctly (third row, the substrate of the denoised image in (b) is missing the streaking pattern that descends diagonally from left to right in (d)). The reason is that the training set is generated using a relatively wide range of imaging parameters, which results in very diverse CeO_2 substrates, as illustrated by the two training examples shown in (b). Training the supervised denoiser on a subset of the data, chosen to align more closely with the real imaging parameters, solves the problem. The substrate in the training examples is more similar to the true one, which enables the denoiser to recover the correct pattern (third row of (c), compare to (d)).